Online Updating for Linear Panel Regressions

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Abstract

In this paper, we develop online updating methods for linear panel regression models. Online updating refers to procedures for sequentially updating parameter estimates as new data become available. In practice, the potential size of the dataset or data confidentiality constraints may preclude researchers from storing or accessing the entire dataset. We propose an online updating procedure for widely used linear regression models in panel data, where data expansion can occur through either (i) the arrival of new units or (ii) the arrival of additional time periods for existing units. The proposed procedure yields closed-form expressions for updating both the point estimates and associated standard errors in each scenario.

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1 Introduction

In the age of big data, computational efficiency has become a central concern in applied econometrics. Researchers increasingly work with administrative, transaction-level, or firm-level datasets containing millions or even billions of observations, often under tight constraints on computation, storage, and data access. In such environments, storing the full dataset on a single machine may be infeasible, and even basic procedures such as ordinary least squares (OLS) estimation can become difficult to implement.

At the same time, the demand for analyzing large and frequently updated datasets continues to grow across many areas of economics and business. Panel data, which combine cross-sectional and time-series dimensions, are widely used to estimate causal effects while controlling for unobserved heterogeneity. However, as either the number of units or the number of time periods increases—and especially when both increase—the size of the dataset grows multiplicatively. This growth can render standard estimation procedures computationally burdensome or infeasible in typical computing environments.

Several recent studies address these challenges by developing memory-efficient algorithms for estimating high-dimensional fixed effects models, often relying on projection methods that avoid direct matrix inversion. While these approaches improve computational scalability, they typically assume access to the full dataset and are not designed for sequential or streaming data. In practice, however, new data often arrive over time—for example, additional firms, transactions, or time periods—and estimates must be updated accordingly. Re-estimating models from scratch becomes increasingly impractical as data grow in size, or when privacy or storage constraints prevent retaining historical datasets. These considerations motivate online estimation methods, which update estimates sequentially as new data arrive, in contrast to offline (batch) approaches that require access to the entire dataset.

In this paper, we develop online updating algorithms for standard panel data models, focusing on pooled ordinary least squares (POLS) and linear fixed effects estimators. We derive algebraically exact updating formulas for regression coefficients and associated standard errors that do not require access to previously observed data and scale linearly with incoming observations. This scalability makes it feasible to estimate fixed effects models directly on massive panel datasets with billions of observations, even when storing or repeatedly processing

the full dataset is infeasible. To our knowledge, this is the first framework that adapts online updating methods to linear panel data models.

A key challenge arises from the dual-dimensional structure of panel data, which allows for two types of data expansion: (i) the arrival of new units and (ii) the extension of time periods for existing units. Our procedure provides closed-form expressions that accommodate both cases while requiring only low-dimensional summary quantities to be stored. In our two-way fixed effects verification exercise, the proposed online updates reproduce the full-sample refit up to numerical precision while using only a small fraction of the memory required by offline estimation.[1]

To illustrate the motivation for our approach, consider a panel dataset with 100 individuals observed over five time periods. When the full dataset is available, standard statistical software (e.g., the `plm` package in R) can easily estimate a two-way fixed effects (TWFE) model. In many applications, however, data do not arrive all at once. Suppose that data on 20 individuals are initially available, while the remaining observations arrive sequentially. When data arrive incrementally, re-estimating the TWFE model from scratch each time quickly becomes inefficient and cumbersome, especially in large-scale settings. Online updating provides a practical alternative by allowing estimates to be updated incrementally without storing or reprocessing the complete dataset. The proposed approach involves additional computation, but in large-scale settings this cost is small relative to repeated full-sample re-estimation.

Figure 1 illustrates this idea. Starting from an initial sample of 20 individuals, we estimate the TWFE model and plot the coefficient as a triangle, with 95% confidence intervals based on HC0 cluster-robust standard errors. As new individual data arrive sequentially, the solid line traces the updated coefficient estimates. The dashed line indicates the full-sample TWFE estimate, with dotted lines representing its 95% confidence interval. Once all data are incorporated, the online estimates coincide exactly with the full-sample results, illustrating that the proposed method delivers the same inference while avoiding repeated full-data re-estimation.

[1] In the baseline configuration ($N = 10{,}000$, $T = 5$), the final dataset required is about 1.37 MB, whereas the online algorithm stores only a compact set of summary matrices of about 21.1 KB, implying roughly 98% memory savings.

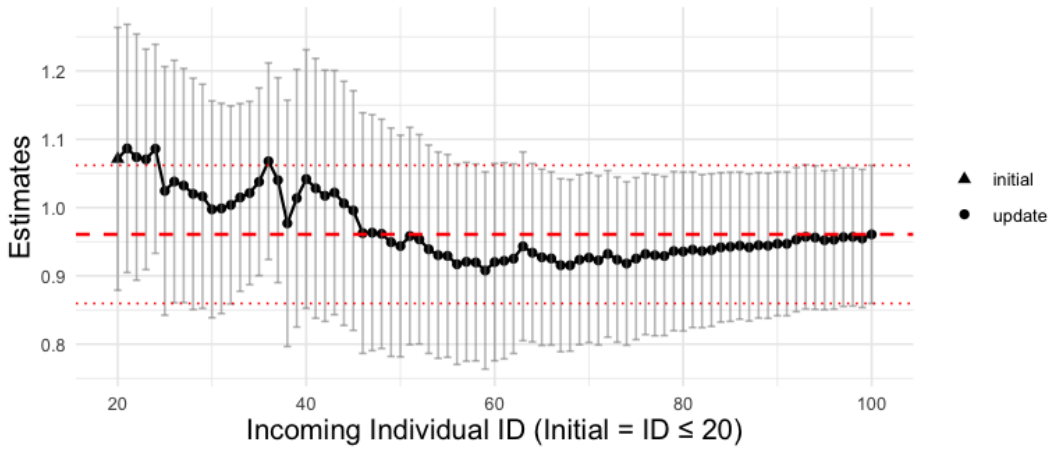

Figure 1: Online updating path of TWFE estimator

1.1 Related Literature

Online updating methods have a long history in cross-sectional econometrics. Early work by Plackett (1950) and Giles (1981) developed recursive formulas for OLS and instrumental variables estimators, while Phillips (1977) provided recursions for two-stage least squares.

Applying online methods to panel data, however, presents additional computational challenges due to the presence of fixed effects. The within-transformation is commonly used to eliminate additive fixed effects in balanced panels, but it is not directly applicable in unbalanced settings (Hansen, 2022). An alternative approach is to include dummy variables for fixed effects, though this substantially increases the dimensionality of the regression design matrix. Somaini & Wolak (2016) proposed a memory-efficient algorithm for estimating two-way fixed effects models in unbalanced panels using the Frisch-Waugh-Lovell (FWL) theorem. By residualizing variables with respect to the fixed effects space, their method enables consistent estimation and inference without explicit matrix inversion. Similarly, Mittag (2019) developed a straightforward algorithm to estimate models with large numbers of fixed effects, with applications in labor economics where firm and match effects are common. While these approaches reduce memory requirements, they still require access to the entire dataset.

Online methods for cross-sectional data have since been extended to broader econometric frameworks. Schifano et al. (2016) developed iterative algorithms for standard error estimation in generalized linear models and proposed a general framework for online updating of parameter estimates and associated inferential tools, including confidence intervals and hypothesis tests.

Their method allows practitioners to conduct valid statistical inference in real time as new data arrive, without storing or reprocessing the full dataset. The online updating literature has also seen substantial development in the context of quantile regression. For example, Chen et al. (2019) developed memory-efficient conditional quantile regression methods, while Jiang & Yu (2024) introduced unconditional quantile regression methods for streaming data, building on Firpo et al. (2009). These methods typically rely on smoothing techniques to facilitate online optimization.

More recent work has emphasized scalable online estimation and inference through stochastic or learning-based updating. Chen et al. (2025b) develops stochastic approximation methods for generalized method of moments (GMM) estimation, in which parameters are updated sequentially using stochastic gradients constructed from incoming data, and establishes valid inference for overidentified models. Relatedly, Leung et al. (2025) proposes an online GMM approach for time series data, while Chen et al. (2025a) develops a stochastic learning framework for overidentified moment-based models using mini-batch updating. Together, these contributions substantially extend online inference to nonlinear and overidentified settings through stochastic or learning-based updates.

A parallel literature in forecasting and high-dimensional regression addresses scalability in streaming settings through data reduction, regularization, or subset-selection strategies. For instance, Adämmer et al. (2025) develops an online forecasting framework that handles massive pools of predictive signals via timevarying coefficients and adaptive subset averaging. Wang et al. (2019) proposes an information-based optimal subdata selection method for big data linear regression, selecting subsets that approximate full-sample estimation. In the context of high-dimensional inference under adaptive sampling, Deshpande et al. (2023) introduces an online debiasing algorithm that corrects for bias arising from both regularization and adaptive sampling. By maintaining summary statistics and applying bias corrections at each update, their method yields asymptotically normal estimators via martingale central limit theorems.

In contrast to these strands of the literature, this paper develops an algebraically exact online updating procedure for linear panel data models with fixed effects. The proposed method sequentially incorporates all incoming observations and yields coefficient and variance estimates that coincide exactly with their offline full-sample counterparts at each update step, without relying on stochastic approximation, data reduction, regularization, or debiasing.

Related work has also interpreted online estimation through alternative lenses. Chen (2023) interprets synthetic control methods as an online learning procedure and provide performance guarantees. Other extensions include survival analysis (Wu et al., 2021), functional data analysis (Yang & Yao, 2023), DSGE models (Cai et al., 2021), and bootstrap methods (Cheng et al., 2024). While our work is conceptually related to reinforcement learning approaches such as multi-armed bandits—where decisions are updated based on sequential feedback—our focus is on parameter estimation and statistical inference rather than decision-making or reward optimization.

The remainder of the paper is structured as follows. Section 2 introduces the concept of online updating using the POLS model as an illustrative case. Section 3 presents the proposed updating procedures for both coefficients and variance estimators in fixed effects models under different data expansion scenarios. Section 4 concludes. All technical derivations are provided in the appendix.

2 Model and Benchmark Estimator

2.1 Data Structure

We consider a panel dataset consisting of N individuals, each observed over T_i $(i = 1, \ldots, N)$ time periods. The panel may be unbalanced, and the total sample size is $n = \sum_{i=1}^{N} T_i = \sum_{t=1}^{T} N_t$, where N_t denotes the number of individuals observed at time t. Throughout, the index t refers to the common calendar time. Let $T := \max\{t : (i,t) \text{ is observed in the pre-update sample}\}$ denote the maximum calendar period in the pre-update sample. For each individual i, let $S_i \subseteq \{1, \cdots, T\}$ be the set of observed calendar periods for individual i. Let S_i be the set of time periods observed for individual i, with cardinality $|S_i| = T_i$, and let S_t be the set of individuals observed at time t, with $|S_t| = N_t$. We use $\otimes$ to denote the Kronecker product and $\text{vec}(\cdot)$ for the vectorization operator.

In standard linear panel data models, fixed effects are treated as nuisance parameters. The one-way fixed effects (OWFE) model includes either individual or time fixed effects, while TWFE model includes both. These fixed effects are typically removed via the within-transformation. For instance, individual fixed effects u_i can be eliminated by subtracting time

means for each individual, regardless of whether the panel is balanced.

However, in the TWFE model with unbalanced data, removing both individual and time effects is more complex. The standard double-demeaning procedure — subtracting individual and time means and adding the total mean — may not fully eliminate the fixed effects when observation patterns differ across individuals and time. The failure of this approach can be illustrated algebraically, as shown below:

$$u_i - \underbrace{\frac{1}{T_i}\sum_{t\in S_i} u_i}_{=u_i} - \frac{1}{N_t}\sum_{i\in S_t} u_i + \underbrace{\frac{1}{n}\sum_{i\in S_t}\sum_{t\in S_i} u_i}_{=n^{-1}\sum_{i\in S_t} T_i u_i} = -\frac{1}{N_t}\sum_{i\in S_t} u_i + \frac{1}{n}\sum_{i\in S_t} T_i u_i \neq 0.$$

To address this issue, we adopt the FWL approach by including time dummies as additional covariates, allowing the TWFE model to be estimated using the OWFE framework (see Hansen, 2022).

Data expansion in panel settings occurs along two dimensions: (i) the addition of a new individual, bringing in a new time series of observations, and (ii) the arrival of additional time periods for an existing individual.

Figure 2 illustrates these two cases using a stylized panel with six individuals observed over up to four time periods. Observed entries are shown as filled cells, while blank cells indicate missing observations. The left panel depicts the addition of a new individual with its associated time series (cells filled with diagonal lines), while the right panel shows the arrival of a previously unobserved time period for an existing individual (cell filled with dots).

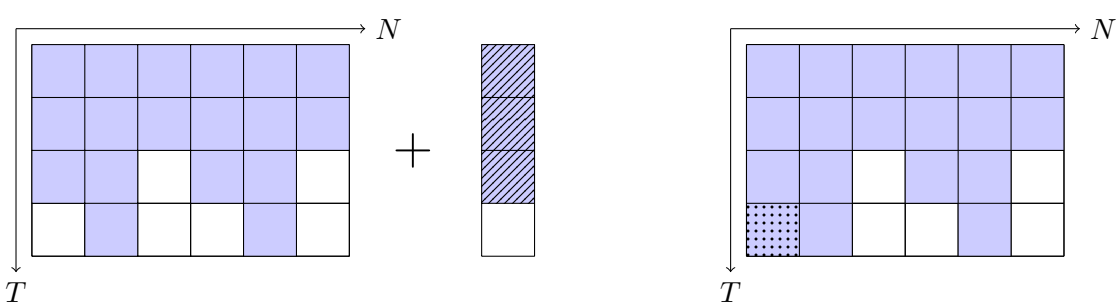

Figure 2: Illustration of panel data expansion. The left panel shows the addition of a new individual. The right panel shows the arrival of a previously missing time period for an existing individual. Blue cells indicate observed data; patterned cells represent newly added observations.

2.2 Pooled OLS

Consider the panel data model:

$$\mathbf{Y}_i = \mathbf{X}_i\beta + \mathbf{e}_i,$$

where $\mathbf{Y}_i$ is a $T_i \times 1$ vector of outcomes, $\mathbf{X}_i$ is a $T_i \times k$ matrix of regressors, and $\mathbf{e}_i$ denotes idiosyncratic errors. Stacking over N individuals yields $\mathbf{Y} = \mathbf{X}\beta + \mathbf{e}$, and the pooled OLS (POLS) estimator is given by

$$\hat{\beta}_{pols} = (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{Y}).$$

Suppose new data arrive in one of two forms: (i) a new individual $(\mathbf{X}_{N+1}, \mathbf{Y}_{N+1})$, or (ii) a new time period $(X_{1,T_1+1}, Y_{1,T_1+1})$ for an existing individual. In either case, the POLS estomator can be updated recursively, without re-estimating the model on the full sample.

Assume that $\mathbf{X}'\mathbf{X}$ is nonsingular. When a new individual $(\mathbf{X}_{N+1}, \mathbf{Y}_{N+1})$ is added, the updated estimator is

$$\hat{\beta}^*_{1,pols} = \hat{\beta}_{pols} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'_{N+1}\left(\mathbf{I} + \mathbf{X}_{N+1}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'_{N+1}\right)^{-1}\left(\mathbf{Y}_{N+1} - \mathbf{X}_{N+1}\hat{\beta}_{pols}\right). \quad (2.1)$$

This update follows directly from the Woodbury matrix identity. In particular, when $\mathbf{X}'\mathbf{X}$ is invertible, the inverse of $\mathbf{X}'\mathbf{X} + \mathbf{X}'_{N+1}\mathbf{X}_{N+1}$ can be expressed as

$$\left(\mathbf{X}'\mathbf{X}\right)^{-1} - \left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'_{N+1}\left(\mathbf{I}_{T_{N+1}} + \mathbf{X}_{N+1}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'_{N+1}\right)^{-1}\mathbf{X}_{N+1}\left(\mathbf{X}'\mathbf{X}\right)^{-1} \quad (2.2)$$

where $\mathbf{I}_{T_{N+1}}$ is the $T_{N+1} \times T_{N+1}$ identity matrix.

For case (ii), where a single new observation arrives for an existing individual, the update reduces to a rank-one correction:

$$\hat{\beta}^*_{2,pols} = \hat{\beta}_{pols} + \frac{(\mathbf{X}'\mathbf{X})^{-1}X_{1,T_1+1}}{1 + X'_{1,T_1+1}(\mathbf{X}'\mathbf{X})^{-1}X_{1,T_1+1}}\left(Y_{1,T_1+1} - X'_{1,T_1+1}\hat{\beta}_{pols}\right). \quad (2.3)$$

Both update formulas depend only on the current coefficient estimate and the stored inverse $(\mathbf{X}'\mathbf{X})^{-1}$, which can be updated sequentially as new data arrive. This illustrates the basic mechanics of online updating in a simple regression setting.

Although these results are standard, they provide a useful benchmark and notation for the developments that follow. In the next sections, we extend this logic to fixed effects models, where additional structure is required to account for unobserved heterogeneity.

3 One-way Fixed Effects

Consider the OWFE model with individual fixed effects:

$$Y_{it} = X_{it}\beta + u_i + e_{it},$$

where u_i represents individual fixed effects. Define the time mean for individual i as $\bar{Y}_i = T_i^{-1}\sum_{t\in S_i} Y_{it}$, and similarly for other variables. By subtracting time means and stacking observations over individuals and time, we obtain:

$$\dot{\mathbf{Y}} = \dot{\mathbf{X}}\beta + \dot{\mathbf{e}},$$

where the "dot" indicates within-transformed variables. The OWFE estimator is then given by:

$$\begin{aligned}\hat{\beta}_{fe} &= \left(\dot{\mathbf{X}}'\dot{\mathbf{X}}\right)^{-1}(\dot{\mathbf{X}}'\dot{\mathbf{Y}}) = \left(\sum_{i=1}^{N}\dot{\mathbf{X}}_i'\dot{\mathbf{X}}_i\right)^{-1}\left(\sum_{i=1}^{N}\dot{\mathbf{X}}_i'\dot{\mathbf{Y}}_i\right) \\ &= \left(\sum_{i=1}^{N}\sum_{t\in S_i}(X_{it}-\bar{X}_i)(X_{it}-\bar{X}_i)'\right)^{-1}\left(\sum_{i=1}^{N}\sum_{t\in S_i}(X_{it}-\bar{X}_i)(Y_{it}-\bar{Y}_i)\right)\end{aligned} \tag{3.1}$$

The online updating procedure for this estimator closely parallels that of the POLS case.

3.1 Updating with a New Individual

Let $(\mathbf{Y}_{N+1}, \mathbf{X}_{N+1})$ be the data for a newly observed individual. We compute the within-transformed variables $\dot{\mathbf{Y}}_{N+1}$ and $\dot{\mathbf{X}}_{N+1}$ by subtracting the time mean for that individual. Specifically,

$$\dot{\mathbf{Y}}_{N+1} = \mathbf{Y}_{N+1} - \bar{Y}_{N+1}\mathbf{1}_{T_{N+1}}, \quad \dot{\mathbf{X}}_{N+1} = \mathbf{X}_{N+1} - \bar{\mathbf{X}}_{N+1}\mathbf{1}_{T_{N+1}},$$

where $\bar{Y}_{N+1} = T_{N+1}^{-1}\sum_{t\in S_{N+1}} Y_{N+1,t}$ and similarly for $\bar{\mathbf{X}}_{N+1}$. Note that this transformation uses only the new individual's data.

Assumption 1. *The cross-product matrix $\dot{\mathbf{X}}'\dot{\mathbf{X}}$ is positive definite.*

Assumption 1 ensures that the inverse of $\dot{\mathbf{X}}'\dot{\mathbf{X}}$ is well defined at the initial update step. Once this condition holds for the initial sample, it is preserved throughout the online updating procedure. Specifically, when new data arrive—either through the addition of new individuals or new time periods—the contribution of the incoming observations to the matrix can be expressed as $\dot{\mathbf{X}}'_{N+1}\dot{\mathbf{X}}_{N+1}$ or $\dot{X}_{i,t^*}\dot{X}'_{i,t^*}$, where t^* denotes a newly observed time period. These terms are positive semidefinite for any realization of the data. As a result, each update adds a positive semidefinite matrix to an already positive definite matrix, implying that $\dot{\mathbf{X}}'\dot{\mathbf{X}}$ remains positive definite at every step. Consequently, once Assumption 1 is satisfied initially, the inverse of $\dot{\mathbf{X}}'\dot{\mathbf{X}}$ continues to exist along the entire updating path. This assumption, however, does not guarantee the existence of an inverse in the two-way fixed effects model. We address the TWFE case separately in Section 4.

Given data on the first N individuals, let $(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}$ denote the stored inverse matrix and $\hat{\beta}_{fe}$ the corresponding fixed effects estimator. The OWFE coefficient vector is updated as follows.

Proposition 1 (New individual: OWFE coefficient update)**.** *Suppose Assumption 1 holds and new individual data $(\mathbf{X}_{N+1}, \mathbf{Y}_{N+1})$ become available. Then, the updated fixed effects estimator based on all $N+1$ individuals, denoted by $\hat{\beta}^*_{1,fe}$, is given by*

$$\hat{\beta}^*_{1,fe} = \hat{\beta}_{fe} + \left(\dot{\mathbf{X}}'\dot{\mathbf{X}}\right)^{-1}\dot{\mathbf{X}}'_{N+1}\mathbf{H}^{-1}_{N+1}\tilde{\mathbf{e}}_{N+1}, \tag{3.2}$$

where $\mathbf{H}_{N+1} := \mathbf{I}_{T_{N+1}} + \dot{\mathbf{X}}_{N+1}\left(\dot{\mathbf{X}}'\dot{\mathbf{X}}\right)^{-1}\dot{\mathbf{X}}'_{N+1}$ *and* $\tilde{\mathbf{e}}_{N+1} := \dot{\mathbf{Y}}_{N+1} - \dot{\mathbf{X}}_{N+1}\hat{\beta}_{fe}$.

Note that $\tilde{\mathbf{e}}_{N+1}$ is evaluated at $\hat{\beta}_{fe}$, not $\hat{\beta}^*_{1,fe}$.

Now consider updating the variance estimator. We start from the classical (homoskedastic, serially uncorrelated) variance estimator,

$$\hat{V}^0_{fe} = \hat{\sigma}^2(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}, \quad \hat{\sigma}^2 = (n-N-k)^{-1}\sum_{i=1}^{N}\hat{\mathbf{e}}'_i\hat{\mathbf{e}}_i \tag{3.3}$$

where $\hat{\mathbf{e}}_i := \dot{\mathbf{Y}}_i - \dot{\mathbf{X}}_i\hat{\beta}_{fe}$ are within-residuals. Let $n^* = n + T_{N+1}$ denote the sample size after adding individual $N+1$. The updated estimator is

$$\hat{V}^{0*}_{1,fe} = \hat{\sigma}^{*2}_1\left(\dot{\mathbf{X}}'\dot{\mathbf{X}} + \dot{\mathbf{X}}'_{N+1}\dot{\mathbf{X}}_{N+1}\right)^{-1},$$

where $\hat{\sigma}^{*2}_1 = (n^* - (N+1) - k)^{-1}\sum_{i=1}^{N+1}\hat{\mathbf{e}}^{*\prime}_i\hat{\mathbf{e}}^*_i$ is the updated residual variance computed at the updated coefficient $\hat{\beta}^*_{1,fe}$. Applying the Woodbury identity to avoid inverting the full matrix $(\dot{\mathbf{X}}'\dot{\mathbf{X}} + \dot{\mathbf{X}}'_{N+1}\dot{\mathbf{X}}_{N+1})^{-1}$ yields the following proposition.

Proposition 2 (New individual: OWFE classical variance update)**.** *Suppose Assumption 1 holds and new individual data* $(\dot{\mathbf{X}}_{N+1}, \dot{\mathbf{Y}}_{N+1})$ *become available. Then, the updated classical OWFE variance estimator based on all* $N+1$ *individuals, denoted by* $\hat{V}^{0*}_{1,fe}$*, is*

$$\hat{V}^{0*}_{1,fe} = \hat{\sigma}^{*2}_1\left((\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1} - (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}'_{N+1}\mathbf{H}^{-1}_{N+1}\dot{\mathbf{X}}_{N+1}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\right), \tag{3.4}$$

where

$$\hat{\sigma}^{*2}_1 := \frac{n-N-k}{n^*-(N+1)-k}\hat{\sigma}^2 + \frac{\tilde{\mathbf{e}}'_{N+1}\mathbf{H}^{-1}_{N+1}\tilde{\mathbf{e}}_{N+1}}{n^*-(N+1)-k}. \tag{3.5}$$

Relative to coefficient updating, variance updating additionally requires updating the residual sum of squares, because $\hat{\sigma}^{*2}_1$ is evaluated at the updated coefficient $\hat{\beta}^*_{1,fe}$. Substituting (3.2) into the definition of $\hat{\mathbf{e}}^*_i$ yields a representation of the updated residuals in terms of the stored inverse $(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}$ and the new individuals within-transformed data, which in turn delivers a

closed-form update for $\sum_{i=1}^{N+1} \hat{\mathbf{e}}_i^{*\prime} \hat{\mathbf{e}}_i^*$. In implementation, this requires maintaining $(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}$ and an additional low-dimensional sufficient matrices for the residual sum of squares (equivalently, $\hat{\sigma}^2$); storing the full historical data is unnecessary.

We now turn to the cluster-robust variance estimator, which allows for arbitrary heteroskedasticity and serial correlation within individuals. The cluster-robust estimator is given by

$$\hat{V}_{fe}^{cr} := a_n(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1} \left(\sum_{i=1}^{N} \dot{\mathbf{X}}_i' \hat{\mathbf{e}}_i \hat{\mathbf{e}}_i' \dot{\mathbf{X}}_i \right) (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1},$$

where a_n is a finite-sample degrees-of-freedom correction (e.g., $a_n = N/(N-1)$). For simplicity, we set $a_n = 1$ throughout.

Unlike the classical variance estimator, updating the cluster-robust requires accounting for how newly arriving data interact with the existing cluster-level residual structure. In particular, the update must accommodate both the contribution of the new individual and the fact that residuals are evaluated at the updated coefficient estimate. The following proposition derives an exact update formula for the cluster-robust variance estimator when a single new individual is added.

Proposition 3 (New individual: OWFE cluster-robust variance update)**.** *Suppose Assumption 1 holds and new individual data* $(\dot{\mathbf{X}}_{N+1}, \dot{\mathbf{Y}}_{N+1})$ *become available. Then, the cluster-robust OWFE variance estimator based on all* $N+1$ *individuals admits an exact online update of the form*

$$\hat{V}_{1,fe}^{cr*} = \mathbf{U}_{N+1} \left\{ \hat{V}_{fe}^{cr} + (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1} \left(-\hat{\mathbf{\Omega}}_{N+1}^{(1)} - \hat{\mathbf{\Omega}}_{N+1}^{(1)'} + \hat{\mathbf{\Omega}}_{N+1}^{(2)} + \hat{\mathbf{\Omega}}_{N+1}^{(3)} \right) (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1} \right\} \mathbf{U}_{N+1}', \quad (3.6)$$

where

$$\hat{\mathbf{\Omega}}_{N+1}^{(1)} := \left\{ \sum_{i=1}^{N} \dot{\mathbf{X}}_i' \dot{\mathbf{Y}}_i \operatorname{vec}(\dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i)' - (\hat{\beta}_{fe}' \otimes \mathbf{I}_k) \sum_{i=1}^{N} \operatorname{vec}(\dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i) \operatorname{vec}(\dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i)' \right\} (\mathbf{I}_k \otimes \mathbf{d}_{N+1}),$$

$$\hat{\mathbf{\Omega}}_{N+1}^{(2)} := (\mathbf{d}_{N+1}' \otimes \mathbf{I}_k) \sum_{i=1}^{N} \operatorname{vec}(\dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i) \operatorname{vec}(\dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i)' (\mathbf{I}_k \otimes \mathbf{d}_{N+1}),$$

$$\hat{\mathbf{\Omega}}_{N+1}^{(3)} := \dot{\mathbf{X}}_{N+1}' \mathbf{H}_{N+1}^{-1} \tilde{\mathbf{e}}_{N+1} \tilde{\mathbf{e}}_{N+1}' \mathbf{H}_{N+1}^{-1} \dot{\mathbf{X}}_{N+1},$$

$$\mathbf{U}_{N+1} := \mathbf{I}_k - (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}'_{N+1}\mathbf{H}^{-1}_{N+1}\dot{\mathbf{X}}_{N+1}, \ \textit{and}$$
$$\mathbf{d}_{N+1} := (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}'_{N+1}\mathbf{H}^{-1}_{N+1}\tilde{\mathbf{e}}_{N+1}.$$

Proposition 3 shows that updating the cluster-robust variance involves additional correction terms that capture how the newly added individual interacts with the pre-existing cluster-level residual structure. These terms arise because residuals entering the cluster-robust are evaluated at the updated coefficient estimate and because the variance estimator aggregates cluster-specific quadratic forms. The resulting update preserves consistency under arbitrary within-individual heteroskedasticity and serial correlation.

From an implementation perspective, the update depends only on previously stored objectsnamely $\hat{V}^{cr}_{fe}$, $(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}$, and a small number of moderate dimensional moment matrices together with the new individuals data. As in the classical case, neither the full historical dataset nor past residual vectors need to be stored or recomputed, making the procedure well suited to large-scale and sequential panel data environments.

The online updating procedure summarized in Algorithm 1 ensures that all necessary quantities are sequentially updated as new individual-level data arrive. Once the new estimates and moment matrices are computed, they replace their previous counterparts and serve as inputs for future updates. The bottom part of the algorithm outlines the minimal set of moment matrices that need to be updated and retained for subsequent rounds of updating.

3.1.1 Space complexity

A central advantage of online updating methods is their ability to operate under strict memory constraints. In this subsection, we compare the space complexity of our online updating algorithm with that of the conventional offline approach that stores and reprocesses the full dataset.

For the proposed online algorithm, memory usage is determined by the largest summary object that must be retained during computation. In the cluster-robust case, this object is the matrix $\sum_{i=1}^{N} \mathrm{vec}(\dot{\mathbf{X}}'_i\dot{\mathbf{X}}_i)\,\mathrm{vec}(\dot{\mathbf{X}}'_i\dot{\mathbf{X}}_i)'$, which has dimension $k^2 \times k^2$. As a result, the space complexity of the online algorithm is $\mathcal{O}(k^4)$ and does not depend on the number of individuals

Algorithm 1: OWFE online update for a new individual data

Input: (1) pre-update state $\mathcal{S}_{\text{pre}} := \left(\hat{\beta}_{fe}, \hat{\sigma}^2, \hat{V}^0_{fe}, \hat{V}^{cr}_{fe}, (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}, A_N, B_N, N, n\right)$,
(2) new individual data $(\mathbf{X}_{\text{new}}, \mathbf{Y}_{\text{new}})$,
(3) $A_N := \sum_{i=1}^N \dot{\mathbf{X}}_i'\dot{\mathbf{Y}}_i \ \text{vec}(\dot{\mathbf{X}}_i'\dot{\mathbf{X}}_i)'$ and $B_N := \sum_{i=1}^N \text{vec}(\dot{\mathbf{X}}_i'\dot{\mathbf{X}}_i) \ \text{vec}(\dot{\mathbf{X}}_i'\dot{\mathbf{X}}_i)'$
Output: post-update state estimates

Within transformation
  Convert $(\mathbf{X}_{\text{new}}, \mathbf{Y}_{\text{new}})$ to within-transformed variables $(\dot{\mathbf{X}}_{\text{new}}, \dot{\mathbf{Y}}_{\text{new}})$
  Set $T_{\text{new}} := \text{nrow}(\dot{\mathbf{X}}_{\text{new}})$
Auxiliary quantities
  Compute $\tilde{\mathbf{e}}_{\text{new}} := \dot{\mathbf{Y}}_{\text{new}} - \dot{\mathbf{X}}_{\text{new}}\hat{\beta}_{fe}$
  Construct $(\mathbf{H}_{\text{new}}, \mathbf{U}_{\text{new}}, \mathbf{d}_{\text{new}})$ as in Propositions 1–3
Point estimate update
  Update $\hat{\beta}^*_{fe}$ using (3.2)
Classical variance update
  Update $\hat{\sigma}^{*2}$ and $\hat{V}^{0*}_{fe}$ using (3.4)–(3.5)
Cluster-robust variance update
  Update $\hat{V}^{cr*}_{fe}$ using (3.6)
State update for subsequent arrivals
  Update $(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}$ via the Woodbury identity (2.2)
  Update $A_N \leftarrow A_N + \dot{\mathbf{X}}_{\text{new}}'\dot{\mathbf{Y}}_{\text{new}} \ \text{vec}(\dot{\mathbf{X}}_{\text{new}}'\dot{\mathbf{X}}_{\text{new}})'$
  Update $B_N \leftarrow B_N + \text{vec}(\dot{\mathbf{X}}_{\text{new}}'\dot{\mathbf{X}}_{\text{new}}) \ \text{vec}(\dot{\mathbf{X}}_{\text{new}}'\dot{\mathbf{X}}_{\text{new}})'$
  Update counts $N \leftarrow N + 1$ and $n \leftarrow n + T_{\text{new}}$
  Overwrite $(\hat{\beta}_{fe}, \hat{\sigma}^2, \hat{V}^0_{fe}, \hat{V}^{cr}_{fe}) \leftarrow (\hat{\beta}^*_{fe}, \hat{\sigma}^{*2}, \hat{V}^{0*}_{fe}, \hat{V}^{cr*}_{fe})$
return post-update state $\mathcal{S}_{\text{post}}$

N or time periods T_i.

By contrast, the conventional offline approach requires storing the raw data matrix. When a new individual is added, memory usage grows proportionally with the total number of observations and covariates, yielding space complexity $\mathcal{O}\left((\sum_{i=1}^{N+1} T_i) \cdot k\right)$.

Table 1 illustrates this contrast for representative values of k. For small samples, the online algorithm may require more memory because it stores higher-order summary matrices. However, as the number of individuals or time periods increases, the offline methods memory requirements grow rapidly and soon exceed those of the online approach. This divergence reflects the fact that the online algorithm maintains only fixed-dimensional summary statistics, whereas the offline method must retain the entire expanding dataset.

As k increases, the space complexity of the online algorithm rises accordingly. For example, increasing k from 5 to 10 increases the size of the stored summary matrix from $5^4 = 625$ to $10^4 = 10{,}000$. This highlights that, in high-dimensional settings, memory usage is governed

N, T_i	$n = \sum_{i=1}^{N} T_i$	**Space complexity**
Online Algorithm ($k = 5$)		
Any N, any T_i	—	$\mathcal{O}(5^4 = 625)$
Offline Algorithm ($k = 5$)		
$N = 10$, $T_i = 3$	30	$\mathcal{O}(30 \times 5 = 150)$
$N = 50$, $T_i = 5$	250	$\mathcal{O}(250 \times 5 = 1250)$
$N = 100$, $T_i = 10$	10^3	$\mathcal{O}(10^3 \times 5 = 5000)$
$N = 1000$, $T_i = 100$	10^5	$\mathcal{O}(10^5 \times 5 = 500000)$
Online Algorithm ($k = 10$)		
Any N, any T_i	—	$\mathcal{O}(10^4 = 10000)$
Offline Algorithm ($k = 10$)		
$N = 10$, $T_i = 3$	30	$\mathcal{O}(30 \times 10 = 300)$
$N = 50$, $T_i = 5$	250	$\mathcal{O}(250 \times 10 = 2500)$
$N = 100$, $T_i = 10$	10^3	$\mathcal{O}(10^3 \times 10 = 10000)$
$N = 1000$, $T_i = 100$	10^5	$\mathcal{O}(10^5 \times 10 = 1000000)$

Table 1: Comparison of Space Complexity: Online vs. Offline Algorithms ($k = 5$ and $k = 10$)

primarily by the number of covariates rather than the sample size.

In many empirical applications, larger values of k are accompanied by correspondingly larger datasets to ensure reliable estimation. In such settings, the offline methods memory requirements scale linearly with the number of observations and quickly become impractical, while the online algorithms memory footprint remains fixed in N and T_i, depending only on k.

3.2 Updating with a New Time Period

We now consider the case in which a new time period is added for an existing individual. Throughout this subsection, it is important to distinguish two interpretations of a new time period for an existing individual, depending on whether the update occurs within the current calendar-time support or extends it.

The distinction is negligible for OWFE regressions, but becomes crucial once we move to TWFE regressions. In a TWFE specification, the regressor vector includes time dummies indexed by the common calendar time. Therefore, if the update adds an observation at an

already existing calendar period (i.e., within the current support), the parameter dimension is unchanged: the set of time dummies remains the same, and the online update only modifies the within-transformed quantities for the affected individual. In contrast, if the update introduces a genuinely new calendar period beyond the pre-update support, the TWFE model must be augmented by an additional time dummy, so that the number of time fixed effects (and hence the parameter dimension) increases by one. Therefore, our TWFE updating formulas and algorithms treat these two cases separately.

By contrast, under OWFE, there is no need to estimate time dummies to be estimated. Consequently, the dimension-expansion issue does not arise, and the only relevant update for an existing individual is the arrival of an additional observation within the given regressor space. We defer the calendar-time extension case to the TWFE section.

Recall that the OWFE estimator in (3.1) can be written as

$$\begin{aligned}\hat{\beta}_{fe} &= \left(\sum_{i=1}^{N}\sum_{t\in S_i}\left(X_{it}-\bar{X}_i\right)X_{it}'\right)^{-1}\left(\sum_{i=1}^{N}\sum_{t\in S_i}\left(X_{it}-\bar{X}_i\right)Y_{it}\right)\\ &= \left(\sum_{i=1}^{N}\sum_{t\in S_i}X_{it}X_{it}'-\sum_{i=1}^{N}\bar{X}_i\bar{X}_i'T_i\right)^{-1}\left(\sum_{i=1}^{N}\sum_{t\in S_i}X_{it}Y_{it}-\sum_{i=1}^{N}\bar{X}_i\bar{Y}_iT_i\right).\end{aligned}$$

Without loss of generality, suppose that the new observation becomes available for individual 1 at some existing calendar period $t_0 \in \{1,\ldots,T\}$ such that $t_0 \notin S_1$ in the pre-update sample. Let $S_1^* := S_1 \cup \{t_0\}$ and $T_1^* := |S_1^*| = T_1 + 1$. All other individuals' observation sets remain unchanged, i.e., $S_i^* = S_i$ for $i \neq 1$. Thus, only the within-transformed quantities associated with individual 1 are modified by the update. Let S_1^* denote the updated index set for individual 1. The following proposition gives a closed-form update for the OWFE coefficient when a single new time period is added.

Proposition 4 (New time: OWFE coefficients update)**.** *Suppose Assumption 1 holds. Fix individual $i = 1$ and add one new time period so that $T_1^* = T_1 + 1$. Then, the updated fixed effects estimator based on the augmented sample, denoted by $\hat{\beta}_{2,fe}^*$, is given by*

$$\hat{\beta}_{2,fe}^* = \hat{\beta}_{fe} + s_{1,T_1^*}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{x}_{1,T_1^*}\tilde{e}_{1,T_1^*}, \tag{3.7}$$

where $\dot{x}_{1,T_1^*} := X_{1,T_1^*} - \bar{X}_1$, $\dot{y}_{1,T_1^*} := Y_{1,T_1^*} - \bar{Y}_1$, $s_{1,T_1^*} := (\kappa^{-1} + h_{1,T_1^*})^{-1}$, $\kappa := \frac{T_1}{T_1^*}$, $h_{1,T_1^*} := \dot{x}'_{1,T_1^*}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{x}_{1,T_1^*}$, *and* $\tilde{e}_{1,T_1^*} := (\dot{y}_{1,T_1^*} - \dot{x}'_{1,T_1^*}\hat{\beta}_{fe})$

Consequently, coefficient updating in this case requires retaining individual-level summary statistics, such as the current means $\bar{X}_1$ and $\bar{Y}_1$, rather than only aggregate cross-sectional quantities.

Updating the inverse of matrix $\dot{\mathbf{X}}'\dot{\mathbf{X}}$ also requires modifying the contribution of individual 1. Let $\dot{\mathbf{X}}_1^*$ denote the within-transformed regressors for individual 1 after the new observation is added. The updated inverse can be written as

$$(\dot{\mathbf{X}}^{*\prime}\dot{\mathbf{X}}^*)^{-1} = (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1} - \frac{\kappa(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}}{1 + \kappa\dot{x}'_{1,T_1^*}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{x}_{1,T_1^*}} = \mathbf{U}_{1,T_1^*}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1},$$

where $\mathbf{U}_{1,T_1^*} := \mathbf{I}_k - s_{1,T_1^*}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*}$. The classical variance estimator is updated as follows:

Proposition 5 (New time: OWFE classical variance update)**.** *Suppose Assumption 1 holds. Fix individual* $i = 1$ *and add one new time period so that* $T_1^* = T_1 + 1$*. Then, the updated classical OWFE variance estimator based on the augmented sample is*

$$\hat{V}_{2,fe}^{0*} = \hat{\sigma}_2^{*2}\mathbf{U}_{1,T_1^*}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}, \tag{3.8}$$

where

$$\hat{\sigma}_2^{*2} = \frac{n - N - k}{n^* - N - k}\hat{\sigma}^2 + \frac{\tilde{e}_{1,T_1^*}^2}{n^* - N - k}\frac{\kappa}{1 + \kappa h_{1,T_1^*}}.$$

Finally, the cluster-robust is updated as follows:

Proposition 6 (New time: OWFE cluster-robust variance update)**.** *Suppose Assumption 1 holds. Fix individual* $i = 1$ *and add one new time period so that* $T_1^* = T_1 + 1$*. Then, the*

updated OWFE cluster-robust variance estimator based on the augmented sample is

$$\hat{V}_{2,fe}^{cr*} = \mathbf{U}_{1,T_1^*}\left\{\hat{V}_{fe}^{cr} + (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\left[-\mathbf{\Omega}_{1,T_1^*}^{(1)} - \mathbf{\Omega}_{1,T_1^*}^{(1)'} + \mathbf{\Omega}_{1,T_1^*}^{(2)} + \mathbf{\Omega}_{1,T_1^*}^{(3)} + \mathbf{\Omega}_{1,T_1^*}^{(4)}\right](\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\right\}\mathbf{U}_{1,T_1^*}' \tag{3.9}$$

where

$$\begin{aligned}
\mathbf{\Omega}_{1,T_1^*}^{(1)} =& \tilde{e}_{1,T_1^*}\left\{\sum_{i=1}^{N}\dot{\mathbf{X}}_i'\dot{\mathbf{Y}}_i\,\mathrm{vec}(\dot{\mathbf{X}}_i'\dot{\mathbf{X}}_i)' - (\hat{\beta}_{fe}'\otimes\mathbf{I}_k)\left(\sum_{i=1}^{N}\mathrm{vec}(\dot{\mathbf{X}}_i'\dot{\mathbf{X}}_i)\,\mathrm{vec}(\dot{\mathbf{X}}_i'\dot{\mathbf{X}}_i)'\right)\right\}\left(\mathbf{I}_k\otimes s_{1,T_1^*}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{x}_{1,T_1^*}\right)\\
\mathbf{\Omega}_{1,T_1^*}^{(2)} =& \tilde{e}_{1,T_1^*}^2(s_{1,T_1^*}\dot{x}_{1,T_1^*}'(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\otimes\mathbf{I}_k)\left(\sum_{i=1}^{N}\mathrm{vec}(\dot{\mathbf{X}}_i'\dot{\mathbf{X}}_i)\,\mathrm{vec}(\dot{\mathbf{X}}_i'\dot{\mathbf{X}}_i)'\right)(\mathbf{I}_k\otimes s_{1,T_1^*}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{x}_{1,T_1^*})\\
\mathbf{\Omega}_{1,T_1^*}^{(3)} =& s_{1,T_1^*}\tilde{e}_{1,T_1^*}(\dot{\mathbf{X}}_1'\hat{\mathbf{e}}_1\dot{x}_{1,T_1^*}' + \dot{x}_{1,T_1^*}\hat{\mathbf{e}}_1'\dot{\mathbf{X}}_1),\\
\mathbf{\Omega}_{1,T_1^*}^{(4)} =& s_{1,T_1^*}^2\tilde{e}_{1,T_1^*}^2\left(\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}' - \dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}'(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1 - \dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}'\right).
\end{aligned}$$

When a new observation is added for an existing individual (here $i = 1$), only that individuals within transformation changes because the individual mean $(\bar{X}_1, \bar{Y}_1)$ and cross-products $(\dot{\mathbf{X}}_1'\dot{\mathbf{Y}}_1, \dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1)$ are recomputed. Hence, the online update can be implemented by revising the contribution of individual 1 while keeping all other individual-level contributions unchanged. Proposition 5 shows that the classical variance update requires only the pre-update inverse matrix $(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}$, the pre-update scalar estimate $\hat{\sigma}^2$, and initial coefficient vector $\hat{\beta}_{fe}$ and time mean which are associated with the new within-transformed regressor $\dot{x}_{1,T_1^*}$ and the corresponding residual component $\tilde{e}_{1,T_1^*}$.

Proposition 6 further shows that the cluster-robust variance update can be computed without storing the full panel history, provided we retain a set of additional aggregate matrices: two vectorization-based summary matrices $\sum_{i=1}^{N}\dot{\mathbf{X}}_i'\dot{\mathbf{Y}}_i\,\mathrm{vec}\,(\dot{\mathbf{X}}_i'\dot{\mathbf{X}}_i)'$ and $\sum_{i=1}^{N}\mathrm{vec}\,(\dot{\mathbf{X}}_i'\dot{\mathbf{X}}_i)\,\mathrm{vec}\,(\dot{\mathbf{X}}_i'\dot{\mathbf{X}}_i)'$ that appear in $\mathbf{\Omega}_{1,T_1^*}^{(1)}$ and $\mathbf{\Omega}_{1,T_1^*}^{(2)}$ and cross-product of pre-update individual 1's covariate matrix $\dot{\mathbf{X}}_1$. Given these stored objects, the arrival of the new time period for individual 1 only requires recomputing the updated within-transformed block for individual 1 (which determines $\tilde{e}_{1,T_1^*}$ and $\dot{x}_{1,T_1^*}$) and then applying the correction terms. Algorithm 2 summarizes the resulting online procedure when a new time observation arrives.

Algorithm 2: update OWFE: new time observation for individual $j \in \{1, \ldots, N\}$

Input: (1) pre-update state $\mathcal{S}_{\text{pre}} := \left(\hat{\beta}_{fe}, \hat{\sigma}^2, \hat{V}^0_{fe}, \hat{V}^{cr}_{fe}, (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}, A_N, B_N, N, n\right)$,

(2) individual j stored summaries $\left(\bar{\mathbf{X}}_j, \bar{Y}_j, T_j,\ \dot{\mathbf{X}}'_j\dot{\mathbf{Y}}_j,\ \dot{\mathbf{X}}'_j\dot{\mathbf{X}}_j\right)$ and incoming observation $(X_{j,t_{\text{new}}}, Y_{j,t_{\text{new}}})$,

(3) $A_N := \sum_{i=1}^N \dot{\mathbf{X}}'_i\dot{\mathbf{Y}}_i \ \text{vec}(\dot{\mathbf{X}}'_i\dot{\mathbf{X}}_i)'$, $\quad B_N := \sum_{i=1}^N \text{vec}(\dot{\mathbf{X}}'_i\dot{\mathbf{X}}_i)\,\text{vec}(\dot{\mathbf{X}}'_i\dot{\mathbf{X}}_i)'$

Output: post-update state and estimates

Update individual means

- $T^*_j \leftarrow T_j + 1$
- $\bar{X}^*_j \leftarrow (T_j\bar{X}_j + X_{j,t_{\text{new}}})/T^*_j$
- $\bar{Y}^*_j \leftarrow (T_j\bar{Y}_j + Y_{j,t_{\text{new}}})/T^*_j$

Compute new within quantities

- Compute $\dot{x}_{j,t_{\text{new}}} := X_{j,t_{\text{new}}} - \bar{X}_j$ and $\dot{y}_{j,t_{\text{new}}} := Y_{j,t_{\text{new}}} - \bar{Y}_j$
- Compute $\kappa := T_j/T^*_j$ and $h_{j,t_{\text{new}}} := \dot{x}'_{j,t_{\text{new}}}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{x}_{j,t_{\text{new}}}$
- Compute $s_{j,t_{\text{new}}} := (\kappa^{-1} + h_{j,t_{\text{new}}})^{-1}$ and $\tilde{e}_{j,t_{\text{new}}} := \dot{y}_{j,t_{\text{new}}} - \dot{x}'_{j,t_{\text{new}}}\hat{\beta}_{fe}$

Coefficient update

- Update $\hat{\beta}^*_{fe}$ using (3.7)

Classical variance update

- Update $\hat{\sigma}^{*2}$ and $\hat{V}^{0*}_{fe}$ using (3.8)

Cluster-robust variance update

- Update $\hat{V}^{cr*}_{fe}$ using (3.9)

Update stored state

- Set $s_j \leftarrow \dot{\mathbf{X}}'_j\dot{\mathbf{Y}}_j$ and $S_j \leftarrow \dot{\mathbf{X}}'_j\dot{\mathbf{X}}_j$
- Update $(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1} \leftarrow \mathbf{U}_{j,T^*_j}\,(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}$
- Update $S^*_j \leftarrow S_j + \kappa\,\dot{x}_{j,t_{\text{new}}}\dot{x}'_{j,t_{\text{new}}}$
- Update $s^*_j \leftarrow s_j + \kappa\,\dot{x}_{j,t_{\text{new}}}\dot{y}_{j,t_{\text{new}}}$
- Update $A_N \leftarrow A_N - s_j\ \text{vec}(S_j)' + s^*_j\ \text{vec}(S^*_j)'$
- Update $B_N \leftarrow B_N - \text{vec}(S_j)\,\text{vec}(S_j)' + \text{vec}(S^*_j)\,\text{vec}(S^*_j)'$
- Store $(\bar{\mathbf{X}}_j, \bar{Y}_j, T_j) \leftarrow (\bar{\mathbf{X}}^*_j, \bar{Y}^*_j, T^*_j)$
- Store $(\dot{\mathbf{X}}'_j\dot{\mathbf{Y}}_j, \dot{\mathbf{X}}'_j\dot{\mathbf{X}}_j) \leftarrow (s^*_j, S^*_j)$
- Update $n \leftarrow n + 1$
- Overwrite $(\hat{\beta}_{fe}, \hat{\sigma}^2, \hat{V}^0_{fe}, \hat{V}^{cr}_{fe}) \leftarrow (\hat{\beta}^*_{fe}, \hat{\sigma}^{*2}, \hat{V}^{0*}_{fe}, \hat{V}^{cr*}_{fe})$

return post-update state $\mathcal{S}_{\text{post}}$

3.3 Space complexity

Algorithm 2 has the same space complexity as Algorithm 1, namely $\mathcal{O}(k^4)$. This is dominated by the $k^2 \times k^2$ summary matrix needed for the cluster-robust variance update. Although individual-level data must be temporarily stored when new time periods arrive, this does not affect the asymptotic storage requirement, as such storage is not accumulated across individuals.

4 Two-way Fixed Effects

Consider the TWFE panel data model with individual and time effects:

$$Y_{it} = X_{it}'\beta + u_i + \tau_t' v + \epsilon_{it},$$

where τ_t is a $T \times 1$ time dummy vector with the tth element equal to 1 and all others 0, and v is a vector of time fixed effects. The product $\tau_t' v$ is written as v_t. In balanced panel data, the within-transformation can be applied to remove individual and time fixed effects. However, in unbalanced panel data, applying the within-transformation is not feasible as explained in the subsection 2.1.

Based on the equivalence between the least squares dummy variable (LSDV) estimator and the fixed effects estimator, one can estimate the TWFE coefficients by applying the within-transformation to remove individual fixed effects and then including time dummies explicitly. The model becomes

$$\dot{\mathbf{Y}} = \dot{\mathbf{X}}\beta + \dot{\mathbf{D}}v + \dot{\boldsymbol{\epsilon}}, \tag{4.1}$$

where $\dot{\mathbf{D}}$ is the stacked matrix of within-transformed time dummies. Absorbing the time dummies into the regressors and applying the earlier updating procedures yields the updated TWFE coefficients and variances.

The TWFE specification includes two sets of regressors, $\dot{\mathbf{X}}$ and $\dot{\mathbf{D}}$. Hence, in contrast to the OWFE case, we must estimate not only the coefficients on $\dot{\mathbf{X}}$, but also the time fixed effects v. Since we consider the TWFE model with unbalanced panel data, the time fixed effects cannot be eliminated by within-transformation. Instead, we control for them by estimating v explicitly.

A key complication is that the number of time effects to be estimated grows whenever a new time period enters the sample. One simple way to avoid this would be to fix in advance a maximal time horizon and declare that only data up to that horizon will ever be used. For instance, the researcher may decide to restrict attention to at most five time periods. If the current dataset already contains at least one individual observed over all five periods, then

TWFE estimation with a fixed set of time dummies is feasible. This strategy, however, has two important limitations: (i) the time horizon must be fixed ex-ante, regardless of how much additional data become available in the future, and (ii) the current dataset must already include at least one individual with observations at all periods up to the maximal horizon.

An alternative is to choose a maximal time horizon that is sufficiently large to accommodate a potentially long stream of future periods. Let $\dot{\mathbf{Z}} := (\dot{\mathbf{X}}, \dot{\mathbf{D}})$ denote the combined regressor matrix. For TWFE estimation, we require $\dot{\mathbf{Z}}'\dot{\mathbf{Z}}$ to be invertible. However, if the maximal time horizon is set much larger than the number of time periods actually present in the current data, then the corresponding time dummies are identically zero for all units in the sample; consequently, many columns of $\dot{\mathbf{D}}$ (and hence of $\dot{\mathbf{Z}}$) are zero, $\dot{\mathbf{Z}}$ fails to have full column rank, and $\dot{\mathbf{Z}}'\dot{\mathbf{Z}}$ becomes singular, so the standard inverse $(\dot{\mathbf{Z}}'\dot{\mathbf{Z}})^{-1}$ is not well-defined. To avoid this issue, we do not fix a maximal time horizon that exceeds the set of time periods observed in the current sample. Instead, we consider an online updating scheme with two cases, depending on whether the incoming observation arrives at an already existing time period or introduces a new time period.

Let the regressors for TWFE as

$$Z_{it} := \left(X_{it}', D_{it}^2, \cdots, D_{it}^T\right)' \in \mathbb{R}^p, \quad p := k + (T-1),$$

where $X_{it} \in \mathbb{R}^k$ and D_{it}^s is the time dummy for calendar period s (excluding a baseline period). Define the individual averages by $\bar{Z}_i := \frac{1}{T_i}\sum_{t \in S_i} Z_{it}$ and $\bar{Y}_i := \frac{1}{T_i}\sum_{t \in S_i} Y_{it}$. The one-way within-transformed variables are defined by $\dot{Z}_{it} := Z_{it} - \bar{Z}_i$ and $\dot{Y}_{it} := Y_{it} - \bar{Y}_i$ for $t \in S_i$. Whenever $\dot{\mathbf{Z}}'\dot{\mathbf{Z}}$ is invertible, the TWFE coefficient vector $\theta := (\beta', v')' \in \mathbb{R}^p$ is estimated by $\hat{\theta} = (\dot{\mathbf{Z}}'\dot{\mathbf{Z}})^{-1}\dot{\mathbf{Z}}'\dot{\mathbf{Y}}$.

As we explained in the subsection 3.2, we here consider the case where the incoming observation introduces a time period which is not previously present in the pre-update sample. Then, a new time dummy column must be appended to $\dot{\mathbf{D}}$, increasing the dimension of Z_{it} from p to $p+1$. The case where the incoming observation belongs to a time period already present in the sample is explained in the subsection 3.2, and it is possible to apply the updating algorithms with slightly modified assumption.

Let t_{new} denote the calendar time of the incoming observation. Here, $t_{\text{new}} = T + 1$ (a new calendar period), so the time dummy dimension increases by one. For the updated individual i, $S_i^* := S_i \cup \{t_{\text{new}}\}$ and $T_i^* := |S_i^*| = T_i + 1$.

As before in the OWFE case, suppose a new observation arrives for individual 1 at calendar time t_{new}, with raw data $(Z_{1,t_{\text{new}}}, Y_{1,t_{\text{new}}})$. Let $S_1^* := S_1 \cup \{t_{\text{new}}\}$ and $T_1^* := |S_1^*| = T_1 + 1$ denote the updated set (and number) of observed calendar periods for individual 1. Define the within-transformed variables of the incoming observation with respect to the pre-update individual mean:

$$\dot{z}_{1,t_{\text{new}}} := Z_{1,t_{\text{new}}} - \bar{Z}_1, \quad \dot{y}_{1,t_{\text{new}}} := Y_{1,t_{\text{new}}} - \bar{Y}_1, \quad \kappa := \frac{T_1}{T_1 + 1}.$$

The scalar $\kappa \in (0, 1)$ depends only on the pre-update number of observations for individual 1. After receiving the new observation, the individual means update as

$$\bar{Z}_1^* \leftarrow \bar{Z}_1 + \frac{1}{T_1 + 1}\dot{z}_{1,t_{\text{new}}}, \quad \bar{Y}_1^* \leftarrow \bar{Y}_1 + \frac{1}{T_1 + 1}\dot{y}_{1,t_{\text{new}}}, \quad T_1^* \leftarrow T_1 + 1,$$

where the means are updated because

$$\begin{aligned}
\bar{Z}_1^* =& \frac{1}{T_1^*} \sum_{t \in S_1^*} Z_{1t} = \frac{1}{T_1 + 1} \left(\sum_{t \in S_1} Z_{1t} + Z_{1,t_{\text{new}}} \right) \\
=& \frac{T_1}{T_1 + 1} \bar{Z}_1 + \frac{1}{T_1 + 1} Z_{1,t_{\text{new}}} \\
=& \bar{Z}_1 + \frac{1}{T_1 + 1} \underbrace{\left(Z_{1,t_{\text{new}}} - \bar{Z}_1 \right)}_{\dot{z}_{1,t_{\text{new}}}}.
\end{aligned}$$

The incoming observation arrives at calendar time $t_{\text{new}} = T + 1$, so a new time dummy column corresponding to period $T + 1$ is appended. The expanded raw regressor vector for the

incoming observation is

$$\begin{bmatrix} Z_{1,t_{\text{new}}} \\ D^{T+1}_{1,t_{\text{new}}} \end{bmatrix} = \begin{bmatrix} X_{1,t_{\text{new}}} \\ D^{2}_{1,t_{\text{new}}} \\ \vdots \\ D^{T}_{1,t_{\text{new}}} \\ D^{T+1}_{1,t_{\text{new}}} \end{bmatrix} \in \mathbb{R}^{p+1}, \quad D^{T+1}_{1,t_{\text{new}}} = 1.$$

Here, $D^{T+1}_{it} := \mathbb{1}\{t = T+1\}$ is the newly introduced time dummy for the new calendar period $T+1$. Thus, $D^{T+1}_{1,t_{\text{new}}} = 1$ for the incoming observation in Case 1, whereas $D^{T+1}_{it} = 0$ for all pre-update observations with $t \leq T$. Since the new time dummy column did not exist before the update, the pre-update mean vector extended to $\mathbb{R}^{p+1}$ is

$$\begin{bmatrix} \bar{Z}_1 \\ 0 \end{bmatrix} \in \mathbb{R}^{p+1}.$$

Define the incoming within-deviation in the expanded dimension by

$$g := \begin{bmatrix} Z_{1,t_{\text{new}}} \\ D^{T+1}_{1,t_{\text{new}}} \end{bmatrix} - \begin{bmatrix} \bar{Z}_1 \\ 0 \end{bmatrix} = \begin{bmatrix} Z_{1,t_{\text{new}}} - \bar{Z}_1 \\ D^{T+1}_{1,t_{\text{new}}} \end{bmatrix} = \begin{bmatrix} \dot{z}_{1,t_{\text{new}}} \\ 1 \end{bmatrix}. \tag{4.2}$$

The vector g is comprised of the vector $\dot{z}_{1,t_{\text{new}}}$ and the scalar 1. The component $\dot{z}_{1,t_{\text{new}}}$ corresponds to the within-transformed original p-dimensional regressor vector, whereas the scalar 1 arises solely from the newly appended time dummy.

Assumption 2. *The cross-product matrix $\dot{\mathbf{Z}}'\dot{\mathbf{Z}} \in \mathbb{R}^{p\times p}$ is positive definite. Moreover, for the incoming observation, new time dummy takes a nonzero value.*

Assumption 2 is a standard full-rank condition ensuring that the pre-update TWFE estimator is well-defined as Assumption 1 in OWFE case. When a new calendar period is introduced, the regressor dimension expands from p to $p+1$ because of the newly appended time-dummy column. Before adding the incoming observation, the augmented cross-product matrix has the block form $\text{diag}(\dot{\mathbf{Z}}'\dot{\mathbf{Z}}, 0)$ and is therefore only positive semidefinite. The update in Proposition 7

adds a rank-one term $\kappa gg'$ that yields a positive definite matrix for post-update. Under the canonical dummy coding, the new time dummy equals one for the incoming observation and hence loads nontrivially on the new dimension, so this non-degeneracy holds automatically.

Proposition 7 (New time: TWFE coefficients update)**.** *Suppose that Assumption 2 holds. Consider that the incoming observation arrives at calendar time $t_{\text{new}} = T + 1$, so a new time dummy column corresponding to period $T + 1$ is appended. Let $\dot{\mathbf{Z}}^*$ and $\dot{\mathbf{Y}}^*$ denote the post-update within-transformed regressor matrix and outcome vector after appending the new time dummy column and adding the incoming observation $(Z_{1,t_{\text{new}}}, Y_{1,t_{\text{new}}})$. Let $\kappa := T_1/(T_1 + 1)$ and define g as in (4.2). Then,*

$$\dot{\mathbf{Z}}^{*\prime}\dot{\mathbf{Z}}^* = \begin{bmatrix} \dot{\mathbf{Z}}'\dot{\mathbf{Z}} & \mathbf{0}_{p\times 1} \\ \mathbf{0}'_{p\times 1} & 0 \end{bmatrix} + \kappa\, gg'. \tag{4.3}$$

Consequently, $\dot{\mathbf{Z}}^{*\prime}\dot{\mathbf{Z}}^*$ is symmetric positive definite, and its inverse is given by

$$\left(\dot{\mathbf{Z}}^{*\prime}\dot{\mathbf{Z}}^*\right)^{-1} = \begin{bmatrix} (\dot{\mathbf{Z}}'\dot{\mathbf{Z}})^{-1} & -(\dot{\mathbf{Z}}'\dot{\mathbf{Z}})^{-1}\dot{z}_{1,t_{\text{new}}} \\ -\dot{z}'_{1,t_{\text{new}}}(\dot{\mathbf{Z}}'\dot{\mathbf{Z}})^{-1} & \dfrac{1 + \kappa\dot{z}'_{1,t_{\text{new}}}(\dot{\mathbf{Z}}'\dot{\mathbf{Z}})^{-1}\dot{z}_{1,t_{\text{new}}}}{\kappa} \end{bmatrix}. \tag{4.4}$$

Moreover, the updated estimator $\hat{\theta}^* = (\dot{\mathbf{Z}}^{*\prime}\dot{\mathbf{Z}}^*)^{-1}\dot{\mathbf{Z}}^{*\prime}\dot{\mathbf{Y}}^*$ satisfies

$$\hat{\theta}^* = \begin{bmatrix} \hat{\theta} \\ \hat{v}_{T+1} \end{bmatrix}, \quad \hat{v}_{T+1} = \dot{y}_{1,t_{\text{new}}} - \dot{z}'_{1,t_{\text{new}}}\hat{\theta}. \tag{4.5}$$

Proposition 7 shows the coefficient update where the incoming observation arrives at a new calendar period $t_{\text{new}} = T + 1$ and thus introduces an additional time dummy regressor, expanding the parameter dimension from p to $p+1$. From an implementation perspective, this update only requires (i) appending a zero column for the new time dummy to the pre-update within-transformed regressors (so the pre-update cross-product becomes $\text{diag}(\dot{\mathbf{Z}}'\dot{\mathbf{Z}}, 0)$) and (ii) adding the rank-one contribution $\kappa gg'$ generated by the new observation. Given the pre-update inverse $(\dot{\mathbf{Z}}'\dot{\mathbf{Z}})^{-1}$, the updated inverse $(\dot{\mathbf{Z}}^{*\prime}\dot{\mathbf{Z}}^*)^{-1}$ is obtained by simple matrix algebra, and the updated estimator $\hat{\theta}^* = (\dot{\mathbf{Z}}^{*\prime}\dot{\mathbf{Z}}^*)^{-1}\dot{\mathbf{Z}}^{*\prime}\dot{\mathbf{Y}}^*$ has the structure in (4.5): the coefficients on all pre-

existing regressors coincide with their pre-update values, while only the new time fixed effect v_{T+1} is newly computed from the incoming residualized outcome.

For classical variance, define the residual vector $\hat{\boldsymbol{\epsilon}} := \dot{\mathbf{Y}} - \dot{\mathbf{Z}}\hat{\theta}$ and the residual sum of squares $\hat{\boldsymbol{\epsilon}}'\hat{\boldsymbol{\epsilon}} = \sum_{i=1}^{N} \hat{e}_i'\hat{e}_i$. The classical variance estimator is

$$\hat{V}_{fe}^{0} := \hat{\sigma}^2(\dot{\mathbf{Z}}'\dot{\mathbf{Z}})^{-1}, \quad \hat{\sigma}^2 := (n - N - p)^{-1}\hat{\boldsymbol{\epsilon}}'\hat{\boldsymbol{\epsilon}},$$

The classical variance estimator is updated as follows:

Proposition 8 (New time: TWFE classical variance update)**.** *Suppose that Assumption 2 holds. Then, $n^* = n+1$ and $p^* = p+1$, so $n^* - N - p^* = n - N - p$. Moreover, the post-update residual sum of squares is unchanged, hence $\hat{\sigma}^{2*} = \hat{\sigma}^2$. Consequently,*

$$\hat{V}_{twfe}^{0*} = \hat{\sigma}^2(\dot{\mathbf{Z}}^{*\prime}\dot{\mathbf{Z}}^*)^{-1}. \tag{4.6}$$

In particular, combining (4.6) with (4.4) yields

$$\hat{V}_{twfe}^{0*} = \hat{\sigma}^2 \begin{bmatrix} (\dot{\mathbf{Z}}'\dot{\mathbf{Z}})^{-1} & -(\dot{\mathbf{Z}}'\dot{\mathbf{Z}})^{-1}\dot{z}_{1,t_{\text{new}}} \\ -\dot{z}_{1,t_{\text{new}}}'(\dot{\mathbf{Z}}'\dot{\mathbf{Z}})^{-1} & \dfrac{1 + \kappa\dot{z}_{1,t_{\text{new}}}'(\dot{\mathbf{Z}}'\dot{\mathbf{Z}})^{-1}\dot{z}_{1,t_{\text{new}}}}{\kappa} \end{bmatrix}. \tag{4.7}$$

Lastly, the cluster-robust variance estimator is updated as follows:

Proposition 9 (New time: TWFE cluster-robust variance update)**.** *Suppose that Assumption 2 holds. Let $r_i := \dot{\mathbf{Z}}_i'\hat{\mathbf{e}}_i \in \mathbb{R}^p$ and $\mathbf{M} := \sum_{i=1}^{N} r_i r_i' \in \mathbb{R}^{p\times p}$. Define their zero-extensions*

$$\tilde{r}_i := \begin{bmatrix} r_i \\ 0 \end{bmatrix} \in \mathbb{R}^{p+1}, \quad \widetilde{\mathbf{M}} := \begin{bmatrix} \mathbf{M} & 0_{p\times 1} \\ 0_{p\times 1}' & 0 \end{bmatrix} \in \mathbb{R}^{(p+1)\times(p+1)}.$$

Let $\hat{\theta}^*$ be the updated TWFE estimator and define post-update residuals $\hat{\mathbf{e}}_i^* := \dot{\mathbf{Y}}_i^* - \dot{\mathbf{Z}}_i^*\hat{\theta}^*$ and $r_i^* := \dot{\mathbf{Z}}_i^{*\prime}\hat{\mathbf{e}}_i^* \in \mathbb{R}^{p+1}$. Under the new-time update for individual 1*, $r_i^* = \tilde{r}_i$ for all $i = 2, \ldots, N$,*

and thus

$$\mathbf{M}^* := \sum_{i=1}^{N} r_i^* r_i^{*'} = \sum_{i=2}^{N} \tilde{r}_i \tilde{r}_i' + r_1^* r_1^{*'} = \widetilde{\mathbf{M}} - \tilde{r}_1 \tilde{r}_1' + r_1^* r_1^{*'}.$$

Consequently, the updated cluster-robust variance estimator is

$$\hat{V}_{twfe}^{cr*} = (\dot{\mathbf{Z}}^{*'} \dot{\mathbf{Z}}^*)^{-1} \mathbf{M}^* (\dot{\mathbf{Z}}^{*'} \dot{\mathbf{Z}}^*)^{-1}. \tag{4.8}$$

Proposition 8 describes the classical variance update. Although the cross product matrix changes, the degrees of freedom $n^* - N - p^*$ remain equal to $n - N - p$, and the residual sum of squares is unchanged in this case, implying $\hat{\sigma}^{*2} = \hat{\sigma}^2$. Therefore, the classical variance update reduces to combining the invariant scalar $\hat{\sigma}^2$ with the updated inverse matrix.

Proposition 9 provides the corresponding cluster-robust variance update. The main implementation challenge is updating the pre-update cluster-level quadratic-form sum $\mathbf{M} = \sum_{i=1}^{N} r_i r_i'$ to account for two complications: the dimension expansion from p to $p+1$, and the fact that, when a new calendar time observation is appended, only individual 1's within-transformed residual is affected. The first change is handled by the zero-extension $\widetilde{\mathbf{M}}$. The second change is handled by removing the pre-update contribution of individual 1 and adding the updated contribution based on the post-update quantities, yielding the rank-one correction

$$\mathbf{M}^* = \widetilde{\mathbf{M}} - \tilde{r}_1 \tilde{r}_1' + r_1^* r_1^{*'}.$$

Moreover, under the updated coefficient $\hat{\theta}^*$, the updated individual 1's contribution admits the compact form

$$r_1^* = \tilde{r}_1 + \kappa g \tilde{e}_{1,t_{\text{new}}} - (\tilde{S}_1 + \kappa g g')(\hat{\theta}^* - \tilde{\theta}),$$

where $S_1 := \dot{\mathbf{Z}}_1' \dot{\mathbf{Z}}_1$ and $\tilde{S}_1 := \begin{bmatrix} S_1 & 0 \\ 0 & 0 \end{bmatrix}$. Thus, $\mathbf{M}^*$ can be computed from stored summaries for individual 1, together with $(\kappa, g, \tilde{e}_{1,t_{\text{new}}})$ obtained from the incoming observation.

One might wonder whether the same strategy extends to the arrival of an arbitrary new regressor. Note that the additional regressor in our setup is a new calendar time dummy, which is identically zero in the pre-update sample. As a result, all pre-update cross products involving the new regressor are known to be zero ($\dot{Z}_{it} D_{i,t_{\text{new}}}^{T+1} = 0$, $D_{i,t_{\text{new}}}^{T+1\prime} D_{i,t_{\text{new}}}^{T+1} = 0$, and $D_{i,t_{\text{new}}}^{T+1\prime} \dot{Y}_{it} = 0$ for $t \leq T$), so the pre-update objects can be embedded into the expanded $(p+1)$-dimensional space by simple zero-extension without any loss of information. This property generally fails for an arbitrary new regressor that takes nonzero values in the pre-update sample. Exact updating would then require the cross products between the newly added regressor and the previously observed regressors and outcome, computed over the entire pre-update sample. These quantities cannot be reconstructed from the pre-update inverse and coefficient estimates unless the historical values of the new regressor have been retained. Thus, the present dimension-expansion update is feasible because the newly introduced time dummy carries no pre-update variation.

Algorithm 3: update TWFE: new calendar period $t_{\text{new}} = T + 1$ for individual $j \in \{1, \ldots, N\}$

Input: (1) pre-update state $\mathcal{S}_{\text{pre}} := \left((\dot{\mathbf{Z}}'\dot{\mathbf{Z}})^{-1}, \hat{\theta}, \hat{\sigma}^2, \hat{V}_{twfe}^{0}, \hat{V}_{twfe}^{cr}, \mathbf{M}\right)$,

(2) individual j stored summaries $\left(\bar{Z}_j, \bar{Y}_j, T_j, \dot{\mathbf{Z}}_j'\dot{\mathbf{Y}}_j, \dot{\mathbf{Z}}_j'\dot{\mathbf{Z}}_j\right)$ and incoming observation $(Z_{j,t_{\text{new}}}, Y_{j,t_{\text{new}}})$

Output: post-update estimates state and estimates

Compute new within quantities

Compute $\dot{z}_{j,t_{\text{new}}}$ and $\dot{y}_{j,t_{\text{new}}}$

Construct $g \in \mathbb{R}^{p+1}$

Compute $\kappa := \frac{T_j}{T_j+1}$

Coefficient update

Update $\hat{\theta}^*$ using (4.5)

Classical variance update

Update $\hat{V}_{twfe}^{0*}$ using (4.7)

Cluster-robust variance update

Update $\hat{V}_{twfe}^{cr*}$ using (4.8)

Update stored state

Update $(\dot{\mathbf{Z}}'\dot{\mathbf{Z}})^{-1} \leftarrow (\dot{\mathbf{Z}}^{*\prime}\dot{\mathbf{Z}}^*)^{-1}$

Update $\dot{\mathbf{Z}}_j'\dot{\mathbf{Z}}_j \leftarrow \begin{bmatrix} \dot{\mathbf{Z}}_j'\dot{\mathbf{Z}}_j & 0 \\ 0 & 0 \end{bmatrix} + \kappa\, gg'$

Update $\dot{\mathbf{Z}}_j'\dot{\mathbf{Y}}_j \leftarrow \begin{bmatrix} \dot{\mathbf{Z}}_j'\dot{\mathbf{Y}}_j \\ 0 \end{bmatrix} + \kappa\, g\, \dot{y}_{j,t_{\text{new}}}$

Compute $\mathbf{M}^* := \widetilde{\mathbf{M}} - \tilde{r}_j\tilde{r}_j' + r_j^* r_j^{*\prime}$

Update $\mathbf{M} \leftarrow \mathbf{M}^*$

Overwrite $(\hat{\theta}, \hat{V}_{twfe}^{0}, \hat{V}_{twfe}^{cr}) \leftarrow (\hat{\theta}^*, \hat{V}_{twfe}^{0*}, \hat{V}_{twfe}^{cr*})$

return post-update state $\mathcal{S}_{\text{post}}$

5 Verification and Simulation

5.1 Vericiation

We use R (version 4.5.2) to validate our theoretical results. We generate a synthetic panel dataset and verify the exactness of the proposed online updating formulas by comparing them with their batch TWFE refits. Compared with the OWFE regression, the TWFE regression incorporates time fixed effects as regressors, and, when a previously unobserved calendar period first appears in the data stream, requires the new-time updating results in Propositions 7-9. Because the TWFE updating procedure mirrors the OWFE case except for the arrival of a new calendar period, we use the TWFE specification to empirically verify Propositions 7–9 in addition to Propositions 1–6.

We consider three hypothetical data-streaming scenarios. First, a new unit arrives with fewer observations than the current maximum calendar period; this case is used to validate Propositions 1–3. Second, the same unit receives an additional observation in the current maximum calendar period; this case is used to validate Propositions 4–6. Third, the unit receives a new observation in a previously unobserved calendar period (i.e., beyond the current maximum period); this case is used to validate Propositions 7–9.

Let $i = 1, \ldots, N$ index units and $t = 1, \ldots, T$ index calendar time. We generate outcomes from the TWFE model

$$Y_{it} = X_{it}'\beta + \alpha_i + \lambda_t + u_{it}, \quad \beta = (\beta_1, \beta_2)'.$$

For each observed pair (i, t), we draw covariates independently as

$$X_{it} = (X_{it,1}, X_{it,2})' \sim \mathcal{N}(0, I_2),$$

unit fixed effects as $\alpha_i \overset{i.i.d.}{\sim} \mathcal{N}\left(0, \sigma_\alpha^2\right)$, and time fixed effects as $\lambda_t \overset{i.i.d.}{\sim} \mathcal{N}\left(0, \sigma_\lambda^2\right)$. Idiosyncratic errors are drawn as $u_{it} \overset{i.i.d.}{\sim} \mathcal{N}\left(0, \sigma_u^2\right)$, independently of $(X_{it}, \alpha_i, \lambda_t)$. To accommodate the new-time event in the third scenario, we draw λ_t for $t = 1, \ldots, T+1$ in advance. In our baseline parameterization, we set $\sigma_\alpha = \sigma_\lambda = \sigma_u = 1$ and fix $\beta = (1.3, -0.7)'$.

Because our theory targets unbalanced panel data, each unit i is observed over a random subset of calendar periods. Specifically, we draw the number of observed periods T_i independently from a discrete uniform distribution on $\{T_{\min}, \ldots, T\}$, where $T_{\min} = 2$ ensures that the within transformation is well-defined. Conditional on T_i, we sample $S_i \subseteq \{1, \ldots, T\}$ uniformly without replacement such that $|S_i| = T_i$, and we retain observations only for $t \in S_i$. This yields an initial sample size $n = \sum_{i=1}^{N} T_i$, which is random and typically smaller than in the balanced panel case. As a mild technical safeguard for constructing time dummies, we additionally require that each calendar period $t \in \{1, \ldots, T\}$ appears at least once in the realized sample.

In all three streaming scenarios, incoming observations are generated from the same data-generating process. In particular, when a new unit arrives (scenario 1), we draw its fixed effect α_{N+1} from the same distribution as α_i, generate its covariates and errors analogously, and form outcomes according to the TWFE equation above. To ensure that the new unit has fewer observations than the current maximum calendar period, we generate its initial observations over a strict subset of pre-existing calendar periods. Specifically, we draw $T_{N+1} \in \{T_{\min}, \cdots, T-1\}$ and then sample $S_{N+1} \subseteq \{1, \cdots, T-1\}$ uniformly without replacement such that $|S_{N+1}| = T_{N+1}$; we retain observations only for $t \in S_{N+1}$. When a new observation arrives in an existing calendar period (scenario 2), we add one observation for the same unit at $t = T$. When a new observation arrives in a new calendar period (scenario 3), we add one observation at $t = T + 1$. In both cases, we generate (X_{it}, u_{it}) and the corresponding Y_{it} in the same manner, using the pre-drawn λ_t.

We set $N = 10{,}000$ and $T = 5$ in the synthetic TWFE design described above. Table 2 reports, for each streaming event and the corresponding proposition, the maximum absolute difference between quantities produced by our online updating algorithms and those obtained from a full-sample TWFE refit using the `plm` package in R. Across all cases, the discrepancies are negligibe: on the order of 10^{-14} for coefficient estimates and at most 10^{-15}–10^{-18} for variance matrices. These magnitudes are consistent with finite-precision floating-point rounding rather than any substantive discrepancy between the two approaches. In particular, since IEEE 754 double-precision arithmetic has machine precision on the order of 10^{-16}, differences at the reported scale are expected when the same estimand is computed via different (but algebraically equivalent) sequences of matrix operations. We therefore conclude that our online

updating formulas reproduce the offline TWFE results up to numerical precision.

Event	Proposition	Quantity	Max.abs.diff
Scenario 1	Prop 1	coefficient	1.55e-14
Scenario 1	Prop 2	classical variance	1.95e-18
Scenario 1	Prop 3	cluster-robust variance	1.52e-18
Scenario 2	Prop 4	coefficient	1.87e-14
Scenario 2	Prop 5	classical variance	1.08e-18
Scenario 2	Prop 6	cluster-robust variance	1.46e-18
Scenario 3	Prop 7	coefficient	2.02e-14
Scenario 3	Prop 8	classical variance	4.22e-15
Scenario 3	Prop 9	cluster-robust variance	1.39e-17

Table 2: Data verification of the online updating algorithms against offline refits.

Tables 3–4 illustrate the storage implications of our updating algorithms when we retain the full set of summary objects required for online updating. Unlike the full-data approach, our method stores only the summary objects described in Algorithms 1–3.

Table 3 varies the unit dimension while holding T fixed. Since the dimension of the within-transformed TWFE regressor is $p = k + (T - 1)$, the dimensions of the key summary matrices are invariant to N. As a result, the storage required for the online state is constant, whereas the raw data size grows linearly with N. This structural difference explains the substantial savings in large-N samples. Conversely, when N is small (e.g., $N = 100$), the stored state matrices can exceed the size of the dataset, yielding negative savings.

Table 4 examines the time dimension while keeping N fixed. Here, the regressor dimension p increases with T, so the matrices needed for cluster-robust updating to grow quickly. Consequently, the storage cost of the online updating rises sharply with T and may eventually exceed the size of the full dataset. This trade-off is inherent to maintaining exact, continuously updatable cluster-robust variance estimates.

N	T	n	online state (KB)	data size (KB)	savings (%)
100	5	375	21.14	15.84	-33.43
500	5	1745	21.14	69.36	69.52
1,000	5	3521	21.14	138.73	84.76
5,000	5	17450	21.14	682.84	96.90
10,000	5	35005	21.14	1368.58	98.46
50,000	5	174760	21.14	6827.76	99.69

Table 3: Memory comparison over the unit dimension (T fixed).

N	T	n	online state (KB)	data size (KB)	savings (%)
1,000	3	2487	13.23	98.34	86.55
1,000	5	3519	30.82	138.66	77.77
1,000	10	6001	191.05	235.61	18.91
1,000	20	11017	1943.69	431.55	-350.40

Table 4: Memory comparison over the time dimension (N fixed).

5.2 Simulation

We conduct simulation to assess the scalability of our algorithms. The objective is to compare their computational speed and memory usage with those of the batch method on a large-scale datasets containing more than one million observations. Specifically, we increase the number of units N sequentially over the set $\{10^6, 5\times10^6, 10^7, 2\times10^7, 5\times10^7, 10^8\}$, and apply the same updating procedure in each case. The number of time periods is fixed at $T = 3$. The DGP follows the same TWFE specification used in the verification. Our main interest is whether the proposed algorithms remain feasible for panel datasets that are too large to be stored in memory, or for which offline estimation is computationally prohibitive.

Prior to updating, we must construct a warm-up sample. This warm-up sample is defined as the smallest dataset for which the matrix $\dot{\mathbf{Z}}'\dot{\mathbf{Z}}$ is positive definite. In our simulation, we start with five units observed over two periods. From this warm-up sample, we construct summary matrices required to implement the algorithms.

To reflect data inflow pattern that may occur in a big data environment, the updating procedure consists of three steps. First, one of the warm-up units, say $i = 1$, receives an observation for a new calander period, $t = 3$. This corresponds to the case in which the newly

arriving time period is not included in the pre-update time support, thereby increasing the parameter dimension. At this step, we apply Algorithm 3. Second, the remaining warm-up units receive observations at $t = 3$. Since $t = 3$ has entered the time support, it is an existing time period, and the update can be carried out using Algorihtm 2. Finally, the remaining units arrive with observation for all time periods. This corresponds to the repeated application of Algorithm 1. Since our focus is on settings with large N, the overall computational cost is mostly driven by the repeated implementation of Algorithm 1.

The measurement of computational time of our online algorithms is performed on the pure update operation time. Excluding data generation, construction, and saving time, we only report updating operation of Algorithms 1-3. However, since Algorithm 1 must process units sequentially, a simple implementation is to measure time by turning a timer on and off for each unit. For example, the overhead accumulates when we execute `proc.time()` in R to determine how much CPU time has taken for the updating with around tens of millions to hundreds of millions observation. Then, the act of timing measurement may be more expensive than the actual update operation, so the measured time may be inflated by the measurement overhead. To avoid this, we measure the computational time for updating in Algorithm 1 at the batch level rather than at the unit level. Importantly, Algorithm 1 itself still process units one by one in the same order and with the same update formulas; only the timing procedure is modified. To be specificy, we partition the incoming new units into consecutive blocks of a fixed size, say 50,000. Within each block, we sequentillay apply the Algorithm 1 update to every unit. We turn on the timer immediately before processing the block and stop it after all updates in the block are completed, recording the elapsed time for that block. Repeating this procedure across all blocks and summing the block-level times yields the total update time for Algorithm 1. This approach replaces one timer invocation per unit with one timer invocation per block, substantially reducing measurement overhead and producing a runtime estimate that more closely reflects the computational cost of the updates themselves.

Table 5 summarizes the simulation results. The reported offline total time includes data frame construction, fitting regression, and variance estimation. Note that to run offline estimation in R, since it requires first to build a data frame, it is inevitable to include construction time. As N increases, the computational required to apply our updating algorithms rises accordingly. For instance, when $N = 10^8$ (300 million observations), the online procedure

N	Online update (sec)	Offline total (sec)	Online storage (MB)	Offline storage (MB)
1,000,000	53.70	21.48	0.02	152.59
5,000,000	281.73	106.08	0.02	762.94
10,000,000	682.73	1056.13	0.02	1525.88
20,000,000	1472.44	–	0.02	–
50,000,000	5088.77	–	0.02	–
100,000,000	9937.77	–	0.02	–

Table 5: Scalability simulation for balanced TWFE regression ($T = 3$).

Notes: Online time reports algorithmic update time only (Algorithms 1–3), excluding data generation/storage. Offline total time includes data-frame construction, `pdata.frame`, `plm` fitting, and variance computation (classical and cluster-robust with `vcovHC`). Dashes ("–") indicate that the corresponding offline step fails due to memory constraints. All experiments were run on a MacBook Air (M2) with 16GB RAM.

completes in about 2.76 hours. In contrast, the offline approach encounters scalability limits. Specifically, at $N = 2 \times 10^7$ (60 million observations), the offline method fails with a "vector memory limit reached" error, indicating that the required intermediate objects exceed the avaiable memory resources in our computing environment.

6 Conclusion

This paper proposes online updating methods for linear panel regressions. Specifically, we show how to update the regression coefficients, classical variance, and cluster-robust variance under two types of panel data expansions. Our results demonstrate that it is not necessary to retain the entire dataset to update the parameters of interest when new data become available. Instead, by storing and updating certain vectors or matrices, the relevant parameters can be efficiently recalculated. These algorithms significantly reduce storage requirements while yielding estimates that are algebraically and numerically identical to those obtained from the full sample.

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A Proofs of Propositions

A.1 Proposition 1

Applying the Woodbury matrix identity to the within transformed variables gives

$$\left(\dot{\mathbf{X}}'\dot{\mathbf{X}} + \dot{\mathbf{X}}'_{N+1}\dot{\mathbf{X}}_{N+1}\right)^{-1} = (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1} - (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}'_{N+1}\mathbf{H}_{N+1}^{-1}\dot{\mathbf{X}}_{N+1}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}.$$

Substituting and using $(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}'\dot{\mathbf{Y}} = \hat{\beta}_{fe}$ yields

$$\hat{\beta}^*_{1,fe} = \hat{\beta}_{fe} + (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}'_{N+1}\Big[\dot{\mathbf{Y}}_{N+1} - \mathbf{H}_{N+1}^{-1}\dot{\mathbf{X}}_{N+1}\hat{\beta}_{fe} - \mathbf{H}_{N+1}^{-1}\dot{\mathbf{X}}_{N+1}\left(\dot{\mathbf{X}}'\dot{\mathbf{X}}\right)^{-1}\dot{\mathbf{X}}'_{N+1}\dot{\mathbf{Y}}_{N+1}\Big]. \tag{A.1}$$

Now use the identity $\mathbf{H}_{N+1}^{-1}\mathbf{H}_{N+1} = \mathbf{I}_{T_{N+1}}$, which implies

$$\mathbf{I}_{T_{N+1}} - \mathbf{H}_{N+1}^{-1}\dot{\mathbf{X}}_{N+1}\left(\dot{\mathbf{X}}'\dot{\mathbf{X}}\right)^{-1}\dot{\mathbf{X}}'_{N+1} = \mathbf{H}_{N+1}^{-1}.$$

Combining terms gives

$$\hat{\beta}^*_{1,fe} = \hat{\beta}_{fe} + (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}'_{N+1}\mathbf{H}_{N+1}^{-1}\big(\dot{\mathbf{Y}}_{N+1} - \dot{\mathbf{X}}_{N+1}\hat{\beta}_{fe}\big),$$

which is (3.2).

A.2 Proposition 2

First, observe that

$$\mathbf{I}_{T_{N+1}} - \dot{\mathbf{X}}_{N+1}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}'_{N+1}\mathbf{H}_{N+1}^{-1} = (\mathbf{H}_{N+1} - \dot{\mathbf{X}}_{N+1}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}'_{N+1})\mathbf{H}_{N+1}^{-1} = \mathbf{H}_{N+1}^{-1}. \tag{A.2}$$

Now the residual $\hat{\mathbf{e}}^*_i$ can be expressed as

$$\hat{\mathbf{e}}^*_i := \dot{\mathbf{Y}}_i - \dot{\mathbf{X}}_i\hat{\beta}^*_{1,fe} \tag{A.3}$$

$$= (\dot{\mathbf{Y}}_i - \dot{\mathbf{X}}_i\hat{\beta}_{fe}) - \dot{\mathbf{X}}_i(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}'_{N+1}\mathbf{H}_{N+1}^{-1}\tilde{\mathbf{e}}_{N+1} \tag{A.4}$$

$$= \begin{cases} \hat{\mathbf{e}}_i - \dot{\mathbf{X}}_i(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}'_{N+1}\mathbf{H}^{-1}_{N+1}\tilde{\mathbf{e}}_{N+1} & \text{for} \quad i = 1, \cdots, N \\ \mathbf{H}^{-1}_{N+1}\tilde{\mathbf{e}}_{N+1} & \text{for} \quad i = N+1. \end{cases} \tag{A.5}$$

Since $\sum_{i=1}^{N} \dot{\mathbf{X}}'_i\hat{\mathbf{e}}_i = 0$, $\sum_{i=1}^{N+1} \hat{\mathbf{e}}_i^{*\prime}\hat{\mathbf{e}}_i^*$ can be written as

$$\begin{aligned} \sum_{i=1}^{N+1} \hat{\mathbf{e}}_i^{*\prime}\hat{\mathbf{e}}_i^* =& \sum_{i=1}^{N} \hat{\mathbf{e}}'_i\hat{\mathbf{e}}_i + \tilde{\mathbf{e}}'_{N+1}\mathbf{H}^{-1}_{N+1}\dot{\mathbf{X}}_{N+1}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}'_{N+1}\mathbf{H}^{-1}_{N+1}\tilde{\mathbf{e}}_{N+1} + \tilde{\mathbf{e}}'_{N+1}\mathbf{H}^{-1}_{N+1}\mathbf{H}^{-1}_{N+1}\tilde{\mathbf{e}}_{N+1} \\ =& \sum_{i=1}^{N} \hat{\mathbf{e}}'_i\hat{\mathbf{e}}_i + \tilde{\mathbf{e}}'_{N+1}\mathbf{H}^{-1}_{N+1}\tilde{\mathbf{e}}_{N+1}. \end{aligned}$$

Therefore,

$$\begin{aligned} \hat{\sigma}_1^{*2} =& \frac{1}{n^* - (N+1) - k}\left\{\sum_{i=1}^{N} \hat{\mathbf{e}}'_i\hat{\mathbf{e}}_i + \tilde{\mathbf{e}}'_{N+1}\mathbf{H}^{-1}_{N+1}\tilde{\mathbf{e}}_{N+1}\right\} \\ =& \frac{n - N - k}{n^* - (N+1) - k}\hat{\sigma}^2 + \frac{\tilde{\mathbf{e}}'_{N+1}\mathbf{H}^{-1}_{N+1}\tilde{\mathbf{e}}_{N+1}}{n^* - (N+1) - k}. \end{aligned}$$

Finally, the updated classical variance estimator is

$$\begin{aligned} \hat{V}^{0*}_{1,fe} =& \hat{\sigma}_1^{*2}(\dot{\mathbf{X}}'\dot{\mathbf{X}} + \dot{\mathbf{X}}'_{N+1}\dot{\mathbf{X}}_{N+1})^{-1} \\ =& \hat{\sigma}_1^{*2}\left((\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1} - (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}'_{N+1}\mathbf{H}^{-1}_{N+1}\dot{\mathbf{X}}_{N+1}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\right). \end{aligned}$$

A.3 Proposition 2

Let $\hat{\mathbf{\Omega}} = \sum_{i=1}^{N} \dot{\mathbf{X}}'_i\hat{\mathbf{e}}_i\hat{\mathbf{e}}'_i\dot{\mathbf{X}}_i$ and $\hat{\mathbf{\Omega}}^* = \sum_{i=1}^{N+1} \dot{\mathbf{X}}'_i\hat{\mathbf{e}}_i^*\hat{\mathbf{e}}_i^{*\prime}\dot{\mathbf{X}}_i$. We use (A.5) to write

$$\begin{aligned} \hat{\mathbf{\Omega}}^* =& \sum_{i=1}^{N} \dot{\mathbf{X}}'_i(\hat{\mathbf{e}}_i - \dot{\mathbf{X}}_i(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}'_{N+1}\mathbf{H}^{-1}_{N+1}\tilde{\mathbf{e}}_{N+1})(\hat{\mathbf{e}}_i - \dot{\mathbf{X}}_i(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}'_{N+1}\mathbf{H}^{-1}_{N+1}\tilde{\mathbf{e}}_{N+1})'\dot{\mathbf{X}}_i \\ &+ \dot{\mathbf{X}}'_{N+1}\mathbf{H}^{-1}_{N+1}\tilde{\mathbf{e}}_{N+1}\tilde{\mathbf{e}}'_{N+1}\mathbf{H}^{-1}_{N+1}\dot{\mathbf{X}}_{N+1} \\ =& \hat{\mathbf{\Omega}} - \sum_{i=1}^{N} \dot{\mathbf{X}}'_i\hat{\mathbf{e}}_i\tilde{\mathbf{e}}'_{N+1}\mathbf{H}^{-1}_{N+1}\dot{\mathbf{X}}_{N+1}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}'_i\dot{\mathbf{X}}_i - \sum_{i=1}^{N} \dot{\mathbf{X}}'_i\dot{\mathbf{X}}_i(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}'_{N+1}\mathbf{H}^{-1}_{N+1}\tilde{\mathbf{e}}_{N+1}\hat{\mathbf{e}}'_i\dot{\mathbf{X}}_i \end{aligned} \tag{A.6}$$

$$+\sum_{i=1}^{N}\dot{\mathbf{X}}_i'\dot{\mathbf{X}}_i(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}_{N+1}'\mathbf{H}_{N+1}^{-1}\tilde{\mathbf{e}}_{N+1}\tilde{\mathbf{e}}_{N+1}'\mathbf{H}_{N+1}^{-1}\dot{\mathbf{X}}_{N+1}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}_i'\dot{\mathbf{X}}_i \tag{A.7}$$

$$+\dot{\mathbf{X}}_{N+1}'\mathbf{H}_{N+1}^{-1}\tilde{\mathbf{e}}_{N+1}\tilde{\mathbf{e}}_{N+1}'\mathbf{H}_{N+1}^{-1}\dot{\mathbf{X}}_{N+1}. \tag{A.8}$$

Using the identity $\operatorname{vec}(ABC) = (C' \otimes A)\operatorname{vec}(B)$, the second term in (A.6) can be written as

$$\begin{aligned}
&\sum_{i=1}^{N}\dot{\mathbf{X}}_i'\hat{\mathbf{e}}_i\tilde{\mathbf{e}}_{N+1}'\mathbf{H}_{N+1}^{-1}\dot{\mathbf{X}}_{N+1}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}_i'\dot{\mathbf{X}}_i\\
=&\sum_{i=1}^{N}\dot{\mathbf{X}}_i'(\dot{\mathbf{Y}}_i-\dot{\mathbf{X}}_i\hat{\beta}_{fe})\tilde{\mathbf{e}}_{N+1}'\mathbf{H}_{N+1}^{-1}\dot{\mathbf{X}}_{N+1}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}_i'\dot{\mathbf{X}}_i\\
=&\sum_{i=1}^{N}\dot{\mathbf{X}}_i'(\dot{\mathbf{Y}}_i-\dot{\mathbf{X}}_i\hat{\beta}_{fe})\operatorname{vec}(\dot{\mathbf{X}}_i'\dot{\mathbf{X}}_i)'\left(\mathbf{I}_k\otimes(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}_{N+1}'\mathbf{H}_{N+1}^{-1}\tilde{\mathbf{e}}_{N+1}\right)\\
=&\left\{\sum_{i=1}^{N}\dot{\mathbf{X}}_i'\dot{\mathbf{Y}}_i\operatorname{vec}(\dot{\mathbf{X}}_i'\dot{\mathbf{X}}_i)'-(\hat{\beta}_{fe}'\otimes\mathbf{I}_k)\left(\sum_{i=1}^{N}\operatorname{vec}(\dot{\mathbf{X}}_i'\dot{\mathbf{X}}_i)\operatorname{vec}(\dot{\mathbf{X}}_i'\dot{\mathbf{X}}_i)'\right)\right\}\\
&\times\left(\mathbf{I}_k\otimes(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}_{N+1}'\mathbf{H}_{N+1}^{-1}\tilde{\mathbf{e}}_{N+1}\right).
\end{aligned}$$

The third term in (A.6) is the transpose of the second term. Similarly, (A.7) can be written as

$$\begin{aligned}
&\sum_{i=1}^{N}\dot{\mathbf{X}}_i'\dot{\mathbf{X}}_i(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}_{N+1}'\mathbf{H}_{N+1}^{-1}\tilde{\mathbf{e}}_{N+1}\tilde{\mathbf{e}}_{N+1}'\mathbf{H}_{N+1}^{-1}\dot{\mathbf{X}}_{N+1}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}_i'\dot{\mathbf{X}}_i\\
=&(\tilde{\mathbf{e}}_{N+1}'\mathbf{H}_{N+1}^{-1}\dot{\mathbf{X}}_{N+1}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\otimes\mathbf{I}_k)\left(\sum_{i=1}^{N}\operatorname{vec}(\dot{\mathbf{X}}_i'\dot{\mathbf{X}}_i)\operatorname{vec}(\dot{\mathbf{X}}_i'\dot{\mathbf{X}}_i)'\right)\\
&\times(\mathbf{I}_k\otimes(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}_{N+1}'\mathbf{H}_{N+1}^{-1}\tilde{\mathbf{e}}_{N+1})
\end{aligned}$$

Combining the results gives $\hat{\mathbf{\Omega}}^* = \hat{\mathbf{\Omega}} - \hat{\mathbf{\Omega}}_{N+1}^{(1)} - \hat{\mathbf{\Omega}}_{N+1}^{(1)'} + \hat{\mathbf{\Omega}}_{N+1}^{(2)} + \hat{\mathbf{\Omega}}_{N+1}^{(3)}$.

Finally, since

$$(\dot{\mathbf{X}}'\dot{\mathbf{X}}+\dot{\mathbf{X}}_{N+1}'\dot{\mathbf{X}}_{N+1})^{-1} = (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1} - (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}_{N+1}'\mathbf{H}_{N+1}^{-1}\dot{\mathbf{X}}_{N+1}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}, \tag{A.9}$$

the updated cluster-robust when new individual data arrive is

$$\begin{aligned}\hat{V}_{1,fe}^{cr*} =& \mathbf{U}_{N+1}\left\{(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\left(\hat{\mathbf{\Omega}} - \hat{\mathbf{\Omega}}_{N+1}^{(1)} - \hat{\mathbf{\Omega}}_{N+1}^{(1)'} + \hat{\mathbf{\Omega}}_{N+1}^{(2)} + \hat{\mathbf{\Omega}}_{N+1}^{(3)}\right)(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\right\}\mathbf{U}_{N+1}' \\ =& \mathbf{U}_{N+1}\left\{\hat{V}_{fe}^{cr} + (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\left(-\hat{\mathbf{\Omega}}_{N+1}^{(1)} - \hat{\mathbf{\Omega}}_{N+1}^{(1)'} + \hat{\mathbf{\Omega}}_{N+1}^{(2)} + \hat{\mathbf{\Omega}}_{N+1}^{(3)}\right)(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\right\}\mathbf{U}_{N+1}'.\end{aligned}$$

A.4 Proposition ??

Observe that

$$\hat{\beta}_{2,fe}^{*} = \left(\sum_{i=1}^{N}\sum_{t\in S_i} X_{it}X_{it}' + X_{1,T_1^*}X_{1,T_1^*}' - \sum_{i=2}^{N}\bar{X}_i\bar{X}_i'T_i - \bar{X}_1^*\bar{X}_1^{*\prime}T_1^*\right)^{-1} \tag{A.10}$$

$$\times\left(\sum_{i=1}^{N}\sum_{t\in S_i} X_{it}Y_{it} + X_{1,T_1^*}Y_{1,T_1^*} - \sum_{i=2}^{N}\bar{X}_i\bar{Y}_iT_i - \bar{X}_1^*\bar{Y}_1^*T_1^*\right), \tag{A.11}$$

where $\bar{X}_1^* := (T_1^*)^{-1}\sum_{t\in S_i^*} X_{1t} = (T_1^*)^{-1}\left(T_1\bar{X}_1 + X_{1,T_1^*}\right)$ and $\bar{Y}_1^*$ is defined similarly. Since

$$\begin{aligned}\bar{X}_1^*\bar{X}_1^{*\prime}T_1^* &= \frac{\left(T_1\bar{X}_1 + X_{1,T_1^*}\right)\left(T_1\bar{X}_1 + X_{1,T_1^*}\right)'}{T_1^*} \\ &= \frac{T_1^2\bar{X}_1\bar{X}_1' + 2T_1X_{1,T_1^*}\bar{X}_1' + X_{1,T_1^*}X_{1,T_1^*}'}{T_1^*} \\ &= T_1\bar{X}_1\bar{X}_1' - \frac{T_1}{T_1^*}\bar{X}_1\bar{X}_1' + \frac{2T_1}{T_1^*}X_{1,T_1^*}\bar{X}_1' + \frac{X_{1,T_1^*}X_{1,T_1^*}'}{T_1^*},\end{aligned}$$

(A.10) can be rearranged as

$$\begin{aligned}&\sum_{i=1}^{N}\sum_{t\in S_i} X_{it}X_{it}' + X_{1,T_1^*}X_{1,T_1^*}' - \sum_{i=2}^{N}\bar{X}_i\bar{X}_i'T_i - \bar{X}_1^*\bar{X}_1^{*\prime}T_1^* \\ &= \sum_{i=1}^{N}\sum_{t\in S_i} X_{it}X_{it}' - \sum_{i=1}^{N}\bar{X}_i\bar{X}_i'T_i + \frac{T_1}{T_1^*}\bar{X}_1\bar{X}_1' - \frac{2T_1}{T_1^*}X_{1,T_1^*}\bar{X}_1' + \frac{T_1}{T_1^*}X_{1,T_1^*}X_{1,T_1^*}' \\ &= \dot{\mathbf{X}}'\dot{\mathbf{X}} + \frac{T_1}{T_1^*}\left(X_{1,T_1^*} - \bar{X}_1\right)\left(X_{1,T_1^*} - \bar{X}_1\right)'.\end{aligned}$$

The inverse is given by

$$\left(\dot{\mathbf{X}}'\dot{\mathbf{X}}\right)^{-1} - \frac{\left(\dot{\mathbf{X}}'\dot{\mathbf{X}}\right)^{-1}\left(X_{1,T_1^*}-\bar{X}_1\right)\left(X_{1,T_1^*}-\bar{X}_1\right)'\left(\dot{\mathbf{X}}'\dot{\mathbf{X}}\right)^{-1}}{\frac{T_1^*}{T_1}+\left(X_{1,T_1^*}-\bar{X}_1\right)'\left(\dot{\mathbf{X}}'\dot{\mathbf{X}}\right)^{-1}\left(X_{1,T_1^*}-\bar{X}_1\right)}, \tag{A.12}$$

by the Sherman-Morrison formula.

Similarly, (A.11) can be rearranged as

$$\begin{aligned}&\sum_{i=1}^{N}\sum_{t\in S_i} X_{it}Y_{it} + X_{1,T_1^*}Y_{1,T_1^*} - \sum_{i=2}^{N}\bar{X}_i\bar{Y}_iT_i - \bar{X}_1^*\bar{Y}_1^*T_1^* \\ &= \dot{\mathbf{X}}'\dot{\mathbf{Y}} + \frac{T_1}{T_1^*}\left(X_{1,T_1^*}-\bar{X}_1\right)\left(Y_{1,T_1^*}-\bar{Y}_1\right).\end{aligned} \tag{A.13}$$

By combining (A.12) and (A.13), the updated OWFE estimator is given by

$$\hat{\beta}_{2,fe}^* = \hat{\beta}_{fe} + \frac{\left(\dot{\mathbf{X}}'\dot{\mathbf{X}}\right)^{-1}\left(X_{1,T_1^*}-\bar{X}_1\right)}{\frac{T_1^*}{T_1}+\left(X_{1,T_1^*}-\bar{X}_1\right)'\left(\dot{\mathbf{X}}'\dot{\mathbf{X}}\right)^{-1}\left(X_{1,T_1^*}-\bar{X}_1\right)}\left\{\left(Y_{1,T_1^*}-\bar{Y}_1\right)-\left(X_{1,T_1^*}-\bar{X}_1\right)'\hat{\beta}_{fe}\right\}.$$

A.5 Updated Inverse when New Time Period Arrive

This subsection is to prove the equation

$$(\dot{\mathbf{X}}^{*\prime}\dot{\mathbf{X}}^*)^{-1} = (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1} - \frac{\kappa(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}'(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}}{1+\kappa\dot{x}_{1,T_1^*}'(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{x}_{1,T_1^*}}.$$

We can rewrite $\dot{\mathbf{X}}^*$ as follows:

$$\underset{T_1^*\times k}{\dot{\mathbf{X}}_1^*} = \begin{bmatrix} X_{1,1}' - \bar{X}_1^{*\prime} \\ X_{1,2}' - \bar{X}_1^{*\prime} \\ \vdots \\ X_{1,T_1}' - \bar{X}_1^{*\prime} \\ X_{1,T_1^*}' - \bar{X}_1^{*\prime} \end{bmatrix}, \quad \bar{X}_1^* = \frac{T_1}{T_1^*}\bar{X}_1 + \frac{1}{T_1^*}X_{1,T_1^*},\ T_1^* \equiv T_1 + 1$$

$$
= \begin{bmatrix} X'_{1,1} - \frac{T_1}{T_1^*}\bar{X}_1 - \frac{1}{T_1^*}X_{1,T_1^*} \\ X'_{1,2} - \frac{T_1}{T_1^*}\bar{X}_1 - \frac{1}{T_1^*}X_{1,T_1^*} \\ \vdots \\ X'_{1,T_1} - \frac{T_1}{T_1^*}\bar{X}_1 - \frac{1}{T_1^*}X_{1,T_1^*} \\ X'_{1,T_1^*} - \frac{T_1}{T_1^*}\bar{X}_1 - \frac{1}{T_1^*}X_{1,T_1^*} \end{bmatrix} = \begin{bmatrix} X'_{1,1} - \bar{X}'_1 + \frac{1}{T_1^*}(\bar{X}'_1 - X'_{1,T_1^*}) \\ X'_{1,2} - \bar{X}'_1 + \frac{1}{T_1^*}(\bar{X}'_1 - X'_{1,T_1^*}) \\ \vdots \\ X'_{1,T_1} - \bar{X}'_1 + \frac{1}{T_1^*}(\bar{X}'_1 - X'_{1,T_1^*}) \\ X'_{1,T_1^*} - \bar{X}'_1 + \frac{1}{T_1^*}(\bar{X}'_1 - X'_{1,T_1^*}) \end{bmatrix}
$$

$$
= \begin{bmatrix} X'_{1,1} - \bar{X}'_1 \\ X'_{1,2} - \bar{X}'_1 \\ \vdots \\ X'_{1,T_1} - \bar{X}'_1 \\ X'_{1,T_1^*} - \bar{X}'_1 \end{bmatrix} + \mathbf{1}_{T_1^*} \cdot \frac{1}{T_1^*}(\bar{X}_1 - X_{1,T_1^*})'
$$

$$
= \begin{bmatrix} \dot{\mathbf{X}}_1 \\ \dot{x}'_{1,T_1^*} \end{bmatrix} - \mathbf{1}_{T_1^*} \cdot \frac{1}{T_1^*}\dot{x}'_{1,T_1^*}.
$$

Then, we can rewrite $\dot{\mathbf{X}}'\dot{\mathbf{X}}$ as follows:

$$
\begin{aligned}
\dot{\mathbf{X}}_1^{*\prime}\dot{\mathbf{X}}_1^* &= \left(\begin{bmatrix} \dot{\mathbf{X}}'_1 & \dot{x}_{1,T_1^*} \end{bmatrix} - \frac{\dot{x}_{1,T_1^*}}{T_1^*} \cdot \mathbf{1}'_{T_1^*}\right)\left(\begin{bmatrix} \dot{\mathbf{X}}_1 \\ \dot{x}'_{1,T_1^*} \end{bmatrix} - \mathbf{1}_{T_1^*} \cdot \frac{\dot{x}'_{1,T_1^*}}{T_1^*}\right) \\
&= \begin{bmatrix} \dot{\mathbf{X}}'_1 & \dot{x}_{1,T_1^*} \end{bmatrix}\begin{bmatrix} \dot{\mathbf{X}}_1 \\ \dot{x}'_{1,T_1^*} \end{bmatrix} - \begin{bmatrix} \dot{\mathbf{X}}'_1 & \dot{x}_{1,T_1^*} \end{bmatrix} \cdot \mathbf{1}_{T_1^*} \cdot \frac{\dot{x}'_{1,T_1^*}}{T_1^*} - \frac{\dot{x}_{1,T_1^*}}{T_1^*} \cdot \mathbf{1}'_{T_1^*} \cdot \begin{bmatrix} \dot{\mathbf{X}}_1 \\ \dot{x}'_{1,T_1^*} \end{bmatrix} + \frac{1}{T_1^*}\dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*} \\
&= \dot{\mathbf{X}}'_1\dot{\mathbf{X}}_1 + \dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*} - \begin{bmatrix} \dot{\mathbf{X}}'_1 & \dot{x}_{1,T_1^*} \end{bmatrix} \cdot \mathbf{1}_{T_1^*} \cdot \frac{\dot{x}'_{1,T_1^*}}{T_1^*} - \frac{\dot{x}_{1,T_1^*}}{T_1^*} \cdot \mathbf{1}'_{T_1^*} \cdot \begin{bmatrix} \dot{\mathbf{X}}_1 \\ \dot{x}'_{1,T_1^*} \end{bmatrix} + \frac{1}{T_1^*}\dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*}
\end{aligned}
$$

Note that

$$
\begin{aligned}
\begin{bmatrix} \dot{\mathbf{X}}'_1 & \dot{x}_{1,T_1^*} \end{bmatrix} \cdot \mathbf{1}_{T_1^*} \cdot \frac{\dot{x}'_{1,T_1^*}}{T_1^*} &= \begin{bmatrix} (X_{1,1} - \bar{X}_1) & \cdots & (X_{1,T_1} - \bar{X}_1) & \dot{x}_{1,T_1^*} \end{bmatrix} \cdot \begin{bmatrix} 1 \\ \vdots \\ 1 \\ 1 \end{bmatrix} \cdot \frac{\dot{x}'_{1,T_1^*}}{T_1^*} \\
&= \left(\sum_{t \in S_1^*} X_{1,t} - T_1^*\bar{X}_1\right) \cdot \frac{\dot{x}'_{1,T_1^*}}{T_1^*}
\end{aligned}
$$

$$
\begin{aligned}
&= \left(T_1\bar{X}_1 + X_{1,T_1^*} - T_1^*\bar{X}_1\right) \cdot \frac{\dot{x}'_{1,T_1^*}}{T_1^*} \\
&= \frac{\dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*}}{T_1^*}.
\end{aligned}
$$

Using this, $\dot{\mathbf{X}}_1^{*'}\dot{\mathbf{X}}_1^*$ can be rewritten as follows:

$$
\begin{aligned}
\dot{\mathbf{X}}_1^{*'}\dot{\mathbf{X}}_1^* =& \dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1 + \dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*} - \frac{\dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*}}{T_1^*} - \frac{\dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*}}{T_1^*} + \frac{1}{T_1^*}\dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*} \\
=& \dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1 + \frac{T_1}{T_1^*}\dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*},
\end{aligned}
$$

where we can get the result after applying the Woodbury identity.

A.6 Proposition 5

For a notational convenience, let $\mathbf{A} := s_{1,T_1^*}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{x}_{1,T_1^*} \equiv \frac{\kappa(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{x}_{1,T_1^*}}{1+\kappa\dot{x}'_{1,T_1^*}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{x}_{1,T_1^*}}$. With augmented sample, the residual evaluated in updated coefficient is defined as follows:

$$
\hat{\mathbf{e}}_i^* = \begin{cases} \dot{\mathbf{Y}}_1^* - \dot{\mathbf{X}}_1^*\hat{\beta}_{2,fe}^* = \dot{\mathbf{Y}}_1^* - \dot{\mathbf{X}}_1^*\hat{\beta}_{fe} - \dot{\mathbf{X}}_1^*\mathbf{A}\tilde{e}_{1,T_1^*}, & \text{for } i = 1 \\ \dot{\mathbf{Y}}_i - \dot{\mathbf{X}}_i\hat{\beta}_{2,fe}^* = \hat{\mathbf{e}}_i - \dot{\mathbf{X}}_i\mathbf{A}\tilde{e}_{1,T_1^*}, & \text{for } i \neq 1 \end{cases}
$$

where the $\hat{\beta}_{2,fe}^* = \hat{\beta}_{fe} + \mathbf{A}\tilde{e}_{1,T_1^*}$ by Proposition 4.

To update the classical variance, we need to update $\hat{\sigma}^2$ to $\hat{\sigma}_2^{*2}$, where $\hat{\sigma}_2^{*2} = \frac{1}{n^*-N-k}\sum_{i=1}^N \hat{\mathbf{e}}_i^{*'}\hat{\mathbf{e}}_i^*$. The sum $\sum_{i=1}^N \hat{\mathbf{e}}_i^{*'}\hat{\mathbf{e}}_i^*$ is divided into two parts: $\hat{\mathbf{e}}_1^{*'}\hat{\mathbf{e}}_1^*$ and $\sum_{i=2}^N \hat{\mathbf{e}}_i^{*'}\hat{\mathbf{e}}_i^*$. We first rewrite $\hat{\mathbf{e}}_1^{*'}\hat{\mathbf{e}}_1^*$. This can be rewritten as

$$
\begin{aligned}
\hat{\mathbf{e}}_1^{*'}\hat{\mathbf{e}}_1^* =& \left(\dot{\mathbf{Y}}_1^* - \dot{\mathbf{X}}_1^*\hat{\beta}_{fe} - \dot{\mathbf{X}}_1^*\mathbf{A}\tilde{e}_{1,T_1^*}\right)' \left(\dot{\mathbf{Y}}_1^* - \dot{\mathbf{X}}_1^*\hat{\beta}_{fe} - \dot{\mathbf{X}}_1^*\mathbf{A}\tilde{e}_{1,T_1^*}\right) \\
=& (\dot{\mathbf{Y}}_1^* - \dot{\mathbf{X}}_1^*\hat{\beta}_{fe})'(\dot{\mathbf{Y}}_1^* - \dot{\mathbf{X}}_1^*\hat{\beta}_{fe}) - (\dot{\mathbf{Y}}_1^* - \dot{\mathbf{X}}_1^*\hat{\beta}_{fe})'\dot{\mathbf{X}}_1^*\mathbf{A}\tilde{e}_{1,T_1^*} \\
& - \tilde{e}_{1,T_1^*}\mathbf{A}'\dot{\mathbf{X}}_1^{*'}(\dot{\mathbf{Y}}_1^* - \dot{\mathbf{X}}_1^*\hat{\beta}_{fe}) + \tilde{e}_{1,T_1^*}\mathbf{A}'\dot{\mathbf{X}}_1^{*'}\dot{\mathbf{X}}_1^*\mathbf{A}\tilde{e}_{1,T_1^*}.
\end{aligned}
$$

First, $(\dot{\mathbf{Y}}_1^* - \dot{\mathbf{X}}_1^*\hat{\beta}_{fe})'(\dot{\mathbf{Y}}_1^* - \dot{\mathbf{X}}_1^*\hat{\beta}_{fe})$ can be written as

$$\dot{\mathbf{Y}}_1^{*\prime}\dot{\mathbf{Y}}_1^* - \dot{\mathbf{Y}}_1^{*\prime}\dot{\mathbf{X}}_1^*\hat{\beta}_{fe} - \hat{\beta}_{fe}'\dot{\mathbf{X}}_1^{*\prime}\dot{\mathbf{Y}}_1^* + \hat{\beta}_{fe}'\dot{\mathbf{X}}_1^{*\prime}\dot{\mathbf{X}}_1^*\hat{\beta}_{fe}.$$

By the subsection A.5, we can rewrite the cross products of $\dot{\mathbf{Y}}_1^*$ or (and) $\dot{\mathbf{X}}_1^*$ as

$$\begin{aligned}
\dot{\mathbf{Y}}_1^{*\prime}\dot{\mathbf{Y}}_1^* =& \dot{\mathbf{Y}}_1'\dot{\mathbf{Y}}_1 + \kappa\dot{y}_{1,T_1^*}^2,\\
\dot{\mathbf{X}}_1^{*\prime}\dot{\mathbf{X}}_1^* =& \dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1 + \kappa\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}',\\
\dot{\mathbf{Y}}_1^{*\prime}\dot{\mathbf{X}}_1^* =& \dot{\mathbf{Y}}_1'\dot{\mathbf{X}}_1 + \kappa\dot{y}_{1,T_1^*}\dot{x}_{1,T_1^*}',
\end{aligned}$$

where $\dot{\mathbf{X}}_1^{*\prime}\dot{\mathbf{Y}}_1^*$ is a transpose of $\dot{\mathbf{Y}}_1^{*\prime}\dot{\mathbf{X}}_1^*$. Using those equations, the cross product of $\dot{\mathbf{Y}}_1^* - \dot{\mathbf{X}}_1^*\hat{\beta}_{fe}$ is

$$\begin{aligned}
&(\dot{\mathbf{Y}}_1'\dot{\mathbf{Y}}_1 + \kappa\dot{y}_{1,T_1^*}^2) - (\dot{\mathbf{Y}}_1'\dot{\mathbf{X}}_1 + \kappa\dot{y}_{1,T_1^*}\dot{x}_{1,T_1^*}')\hat{\beta}_{fe} - \hat{\beta}_{fe}'(\dot{\mathbf{X}}_1'\dot{\mathbf{Y}}_1 + \kappa\dot{x}_{1,T_1^*}\dot{y}_{1,T_1^*}) + \hat{\beta}_{fe}'(\dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1 + \kappa\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}')\hat{\beta}_{fe}\\
=&(\dot{\mathbf{Y}}_1 - \dot{\mathbf{X}}_1\hat{\beta}_{fe})'(\dot{\mathbf{Y}}_1 - \dot{\mathbf{X}}_1\hat{\beta}_{fe}) + \kappa\left(\dot{y}_{1,T_1^*}^2 - \dot{y}_{1,T_1^*}\dot{x}_{1,T_1^*}'\hat{\beta}_{fe} - \hat{\beta}_{fe}'\dot{x}_{1,T_1^*}\dot{y}_{1,T_1^*} + \hat{\beta}_{fe}'\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}'\hat{\beta}_{fe}\right)\\
=&\hat{\mathbf{e}}_1'\hat{\mathbf{e}}_1 + \kappa(\dot{y}_{1,T_1^*} - \dot{x}_{1,T_1^*}'\hat{\beta}_{fe})'(\dot{y}_{1,T_1^*} - \dot{x}_{1,T_1^*}'\hat{\beta}_{fe})\\
=&\hat{\mathbf{e}}_1'\hat{\mathbf{e}}_1 + \kappa\tilde{e}_{1,T_1^*}^2.
\end{aligned}$$

Second, $(\dot{\mathbf{Y}}_1^* - \dot{\mathbf{X}}_1^*\hat{\beta}_{fe})'\dot{\mathbf{X}}_1^*\mathbf{A}\tilde{e}_{1,T_1^*}$ can be rewritten as

$$\begin{aligned}
&\dot{\mathbf{Y}}_1^{*\prime}\dot{\mathbf{X}}_1^*\mathbf{A}\tilde{e}_{1,T_1^*} - \hat{\beta}_{fe}'\dot{\mathbf{X}}_1^{*\prime}\dot{\mathbf{X}}_1^*\mathbf{A}\tilde{e}_{1,T_1^*}\\
=&(\dot{\mathbf{Y}}_1'\dot{\mathbf{X}}_1 + \kappa\dot{y}_{1,T_1^*}\dot{x}_{1,T_1^*}')\mathbf{A}\tilde{e}_{1,T_1^*} - \hat{\beta}_{fe}'(\dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1 + \kappa\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}')\mathbf{A}\tilde{e}_{1,T_1^*}\\
=&(\dot{\mathbf{Y}}_1' - \hat{\beta}_{fe}'\dot{\mathbf{X}}_1')\dot{\mathbf{X}}_1\mathbf{A}\tilde{e}_{1,T_1^*} + \kappa(\dot{y}_{1,T_1^*} - \hat{\beta}_{fe}'\dot{x}_{1,T_1^*})\dot{x}_{1,T_1^*}'\mathbf{A}\tilde{e}_{1,T_1^*}\\
=&\hat{\mathbf{e}}_1'\dot{\mathbf{X}}_1\mathbf{A}\tilde{e}_{1,T_1^*} + \kappa\tilde{e}_{1,T_1^*}^2\dot{x}_{1,T_1^*}'\mathbf{A},
\end{aligned}$$

and $\tilde{e}_{1,T_1^*}\mathbf{A}'\dot{\mathbf{X}}_1^{*\prime}(\dot{\mathbf{Y}}_1^* - \dot{\mathbf{X}}_1^*\hat{\beta}_{fe})$ is a transpose of $(\dot{\mathbf{Y}}_1^* - \dot{\mathbf{X}}_1^*\hat{\beta}_{fe})'\dot{\mathbf{X}}_1^*\mathbf{A}\tilde{e}_{1,T_1^*}$.

Third, $\tilde{e}_{1,T_1^*}\mathbf{A}'\dot{\mathbf{X}}_1^{*\prime}\dot{\mathbf{X}}_1^*\mathbf{A}\tilde{e}_{1,T_1^*}$ can be rewritten as

$$\tilde{e}_{1,T_1^*}\mathbf{A}'(\dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1 + \kappa\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}')\mathbf{A}\tilde{e}_{1,T_1^*}$$

$$=\tilde{e}^2_{1,T_1^*}\mathbf{A}'\dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1\mathbf{A}+\kappa\tilde{e}^2_{1,T_1^*}\mathbf{A}'\dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*}\mathbf{A}.$$

Thus, $\hat{\mathbf{e}}_1^{*\prime}\hat{\mathbf{e}}_1^*$ is

$$\begin{aligned}&(\hat{\mathbf{e}}_1'\hat{\mathbf{e}}_1+\kappa\tilde{e}^2_{1,T_1^*})-\left(\hat{\mathbf{e}}_1'\dot{\mathbf{X}}_1\mathbf{A}\tilde{e}_{1,T_1^*}+\kappa\tilde{e}^2_{1,T_1^*}\dot{x}'_{1,T_1^*}\mathbf{A}\right)\\&-\left(\tilde{e}_{1,T_1^*}\mathbf{A}'\dot{\mathbf{X}}_1'\hat{\mathbf{e}}_1-\kappa\tilde{e}^2_{1,T_1^*}\dot{x}'_{1,T_1^*}\mathbf{A}\right)+\left(\tilde{e}^2_{1,T_1^*}\mathbf{A}'\dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1\mathbf{A}+\kappa\tilde{e}^2_{1,T_1^*}\mathbf{A}'\dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*}\mathbf{A}\right)\end{aligned}$$

We can rewrite $\sum_{i=2}^N\hat{\mathbf{e}}_i^{*\prime}\hat{\mathbf{e}}_i^*$ as follows:

$$\begin{aligned}&\sum_{i=2}^N(\hat{\mathbf{e}}_i-\dot{\mathbf{X}}_i\mathbf{A}\tilde{e}_{1,T_1^*})'(\hat{\mathbf{e}}_i-\dot{\mathbf{X}}_i\mathbf{A}\tilde{e}_{1,T_1^*})\\=&\sum_{i=2}^N\hat{\mathbf{e}}_i'\hat{\mathbf{e}}_i-\sum_{i=2}^N\hat{\mathbf{e}}_i'\dot{\mathbf{X}}_i\mathbf{A}\tilde{e}_{1,T_1^*}-\tilde{e}_{1,T_1^*}\mathbf{A}'\sum_{i=2}^N\dot{\mathbf{X}}_i'\hat{\mathbf{e}}_i+\tilde{e}^2_{1,T_1^*}\mathbf{A}'\sum_{i=2}^N\dot{\mathbf{X}}_i'\dot{\mathbf{X}}_i\mathbf{A}\end{aligned}$$

Combining the results derived above, the matrix $\sum_{i=1}^N\dot{\mathbf{e}}_i^{*\prime}\dot{\mathbf{e}}_i^*$ can be written as follows:

$$\begin{aligned}\sum_{i=1}^N\dot{\mathbf{e}}_i^{*\prime}\dot{\mathbf{e}}_i^*=&\sum_{i=1}^N\hat{\mathbf{e}}_i'\hat{\mathbf{e}}_i+\tilde{e}^2_{1,T_1^*}\mathbf{A}'\dot{\mathbf{X}}'\dot{\mathbf{X}}\mathbf{A}+\kappa\tilde{e}^2_{1,T_1^*}-2\kappa\tilde{e}^2_{1,T_1^*}\mathbf{A}'\dot{x}_{1,T_1^*}+\kappa\tilde{e}^2_{1,T_1^*}(\mathbf{A}'\dot{x}_{1,T_1^*})(\mathbf{A}'\dot{x}_{1,T_1^*})'\\=&\sum_{i=1}^N\hat{\mathbf{e}}_i'\hat{\mathbf{e}}_i+\tilde{e}^2_{1,T_1^*}\mathbf{A}'(\dot{\mathbf{X}}'\dot{\mathbf{X}})\mathbf{A}+\kappa\tilde{e}^2_{1,T_1^*}\left(1-2\mathbf{A}'\dot{x}_{1,T_1^*}+(\mathbf{A}'\dot{x}_{1,T_1^*})^2\right)\\=&\sum_{i=1}^N\hat{\mathbf{e}}_i'\hat{\mathbf{e}}_i+\tilde{e}^2_{1,T_1^*}\mathbf{A}'(\dot{\mathbf{X}}'\dot{\mathbf{X}})\mathbf{A}+\kappa\tilde{e}^2_{1,T_1^*}(1-\mathbf{A}'\dot{x}_{1,T_1^*})^2.\end{aligned}$$

Finally, we can calculate $\hat{\sigma}_2^{*2}$. Before, we revert to the original notation and write $\frac{\kappa(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{x}_{1,T_1^*}}{1+\kappa\dot{x}'_{1,T_1^*}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{x}_{1,T_1^*}}$ in place of $\mathbf{A}$. Thus, we can write the expression as follows:

$$\begin{aligned}\mathbf{A}'(\dot{\mathbf{X}}'\dot{\mathbf{X}})\mathbf{A}=&\frac{\kappa^2\dot{x}'_{1,T_1^*}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{x}_{1,T_1^*}}{(1+\kappa\dot{x}'_{1,T_1^*}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{x}_{1,T_1^*})^2}=s^2_{1,T_1^*}h_{1,T_1^*},\\\mathbf{A}'\dot{x}_{1,T_1^*}=&\frac{\kappa\dot{x}'_{1,T_1^*}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{x}_{1,T_1^*}}{1+\kappa\dot{x}'_{1,T_1^*}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{x}_{1,T_1^*}}=s_{1,T_1^*}h_{1,T_1^*}.\end{aligned}$$

Then,

$$
\begin{aligned}
\sum_{i=1}^{N} \hat{\mathbf{e}}_i^{*\prime} \hat{\mathbf{e}}_i^{*} &= \sum_{i=1}^{N} \hat{\mathbf{e}}_i' \hat{\mathbf{e}}_i + \tilde{e}_{1,T_1^*}^2 s_{1,T_1^*}^2 h_{1,T_1^*} + \kappa \tilde{e}_{1,T_1^*}^2 (1 - s_{1,T_1^*} h_{1,T_1^*})^2 \\
&= \sum_{i=1}^{N} \hat{\mathbf{e}}_i' \hat{\mathbf{e}}_i + \tilde{e}_{1,T_1^*}^2 (s_{1,T_1^*}^2 h_{1,T_1^*} + \kappa (1 - s_{1,T_1^*} h_{1,T_1^*})^2) \\
&= \sum_{i=1}^{N} \hat{\mathbf{e}}_i' \hat{\mathbf{e}}_i + \tilde{e}_{1,T_1^*}^2 \left(s_{1,T_1^*}^2 h_{1,T_1^*} + \frac{\kappa}{(1 + \kappa h_{1,T_1^*})^2} \right) \\
&= \sum_{i=1}^{N} \hat{\mathbf{e}}_i' \hat{\mathbf{e}}_i + \tilde{e}_{1,T_1^*}^2 \frac{\kappa}{1 + \kappa h_{1,T_1^*}}
\end{aligned}
$$

Thus, $\hat{\sigma}_2^{*2}$ is derived as

$$
\begin{aligned}
&\frac{1}{n^* - N - k} \sum_{i=1}^{N} \hat{\mathbf{e}}_i^{*\prime} \hat{\mathbf{e}}_i^{*} \\
=&\frac{1}{n^* - N - k} \left(\sum_{i=1}^{N} \hat{\mathbf{e}}_i' \hat{\mathbf{e}}_i + \tilde{e}_{1,T_1^*}^2 \frac{\kappa}{1 + \kappa h_{1,T_1^*}} \right) \\
=&\frac{n - N - k}{n^* - N - k} \hat{\sigma}^2 + \frac{\tilde{e}_{1,T_1^*}^2}{n^* - N - k} \frac{\kappa}{1 + \kappa h_{1,T_1^*}}.
\end{aligned}
$$

A.7 Proposition 6

As we set in A.6, let $\mathbf{A} := \frac{\kappa (\dot{\mathbf{X}}' \dot{\mathbf{X}})^{-1} \dot{x}_{1,T_1^*}}{1 + \kappa \dot{x}_{1,T_1^*}' (\dot{\mathbf{X}}' \dot{\mathbf{X}})^{-1} \dot{x}_{1,T_1^*}}$ for a notational convenience. Since we already defined how the residual changes when a new time period for unit 1 arrives in A.6, we just need to decompose $\sum_{i=1}^{N} \dot{\mathbf{X}}_i^{*\prime} \hat{\mathbf{e}}_i^* \hat{\mathbf{e}}_i^{*\prime} \dot{\mathbf{X}}_i^* = \dot{\mathbf{X}}_1^{*\prime} \hat{\mathbf{e}}_1^* \hat{\mathbf{e}}_1^{*\prime} \dot{\mathbf{X}}_1^* + \sum_{i=2}^{N} \dot{\mathbf{X}}_i' \hat{\mathbf{e}}_i^* \hat{\mathbf{e}}_i^{*\prime} \dot{\mathbf{X}}_i$. Since $\hat{\mathbf{e}}_1^*$ can be rewritten as $\dot{\mathbf{Y}}_1^* - \dot{\mathbf{X}}_1^* \hat{\beta}_{fe} - \dot{\mathbf{X}}_1^* \mathbf{A} \tilde{e}_{1,T_1^*}$, we can rewrite $\dot{\mathbf{X}}_1^{*\prime} \hat{\mathbf{e}}_1^* \hat{\mathbf{e}}_1^{*\prime} \dot{\mathbf{X}}_1^*$ as follows:

$$
\begin{aligned}
&\dot{\mathbf{X}}_1^{*\prime} (\dot{\mathbf{Y}}_1^* - \dot{\mathbf{X}}_1^* \hat{\beta}_{fe} - \dot{\mathbf{X}}_1^* \mathbf{A} \tilde{e}_{1,T_1^*}) (\dot{\mathbf{Y}}_1^* - \dot{\mathbf{X}}_1^* \hat{\beta}_{fe} - \dot{\mathbf{X}}_1^* \mathbf{A} \tilde{e}_{1,T_1^*})' \dot{\mathbf{X}}_1^* \\
=& \left(\dot{\mathbf{X}}_1^{*\prime} \dot{\mathbf{Y}}_1^* - \dot{\mathbf{X}}_1^{*\prime} \dot{\mathbf{X}}_1^* \hat{\beta}_{fe} - \dot{\mathbf{X}}_1^{*\prime} \dot{\mathbf{X}}_1^* \mathbf{A} \tilde{e}_{1,T_1^*} \right) \left(\dot{\mathbf{X}}_1^{*\prime} \dot{\mathbf{Y}}_1^* - \dot{\mathbf{X}}_1^{*\prime} \dot{\mathbf{X}}_1^* \hat{\beta}_{fe} - \dot{\mathbf{X}}_1^{*\prime} \dot{\mathbf{X}}_1^* \mathbf{A} \tilde{e}_{1,T_1^*} \right)' \\
=& \left[(\dot{\mathbf{X}}_1' \dot{\mathbf{Y}}_1 + \kappa \dot{x}_{1,T_1^*} \dot{y}_{1,T_1^*}) - (\dot{\mathbf{X}}_1' \dot{\mathbf{X}}_1 + \kappa \dot{x}_{1,T_1^*} \dot{x}_{1,T_1^*}') \hat{\beta}_{fe} - (\dot{\mathbf{X}}_1' \dot{\mathbf{X}}_1 + \kappa \dot{x}_{1,T_1^*} \dot{x}_{1,T_1^*}') \mathbf{A} \tilde{e}_{1,T_1^*} \right] \\
&\times \left[(\dot{\mathbf{X}}_1' \dot{\mathbf{Y}}_1 + \kappa \dot{x}_{1,T_1^*} \dot{y}_{1,T_1^*}) - (\dot{\mathbf{X}}_1' \dot{\mathbf{X}}_1 + \kappa \dot{x}_{1,T_1^*} \dot{x}_{1,T_1^*}') \hat{\beta}_{fe} - (\dot{\mathbf{X}}_1' \dot{\mathbf{X}}_1 + \kappa \dot{x}_{1,T_1^*} \dot{x}_{1,T_1^*}') \mathbf{A} \tilde{e}_{1,T_1^*} \right]' \\
=& \left[\dot{\mathbf{X}}_1' \hat{\mathbf{e}}_1 - \dot{\mathbf{X}}_1' \dot{\mathbf{X}}_1 \mathbf{A} \tilde{e}_{1,T_1^*} + \kappa \dot{x}_{1,T_1^*} \tilde{e}_{1,T_1^*} - \kappa \dot{x}_{1,T_1^*} \dot{x}_{1,T_1^*}' \mathbf{A} \tilde{e}_{1,T_1^*} \right]
\end{aligned}
$$

$$
\begin{aligned}
&\times \left[\dot{\mathbf{X}}_1'\hat{\mathbf{e}}_1 - \dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1\mathbf{A}\tilde{e}_{1,T_1^*} + \kappa\dot{x}_{1,T_1^*}\tilde{e}_{1,T_1^*} - \kappa\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}'\mathbf{A}\tilde{e}_{1,T_1^*}\right]' \\
=&\left[\dot{\mathbf{X}}_1'\hat{\mathbf{e}}_1 + \left(\kappa\dot{x}_{1,T_1^*} - (\dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1 + \kappa\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}')\mathbf{A}\right)\tilde{e}_{1,T_1^*}\right] \\
&\times \left[\dot{\mathbf{X}}_1'\hat{\mathbf{e}}_1 + \left(\kappa\dot{x}_{1,T_1^*} - (\dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1 + \kappa\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}')\mathbf{A}\right)\tilde{e}_{1,T_1^*}\right]' \\
=&\dot{\mathbf{X}}_1'\hat{\mathbf{e}}_1\hat{\mathbf{e}}_1'\dot{\mathbf{X}}_1 + \dot{\mathbf{X}}_1'\hat{\mathbf{e}}_1\tilde{e}_{1,T_1^*}\left(\kappa\dot{x}_{1,T_1^*} - (\dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1 + \kappa\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}')\mathbf{A}\right)' + \left(\kappa\dot{x}_{1,T_1^*} - (\dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1 + \kappa\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}')\mathbf{A}\right)\tilde{e}_{1,T_1^*}\hat{\mathbf{e}}_1'\dot{\mathbf{X}}_1 \\
&+ \left(\kappa\dot{x}_{1,T_1^*} - (\dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1 + \kappa\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}')\mathbf{A}\right)\left(\kappa\dot{x}_{1,T_1^*} - (\dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1 + \kappa\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}')\mathbf{A}\right)'\tilde{e}_{1,T_1^*}^2 \\
=&\dot{\mathbf{X}}_1'\hat{\mathbf{e}}_1\hat{\mathbf{e}}_1'\dot{\mathbf{X}}_1 + \kappa\tilde{e}_{1,T_1^*}\dot{\mathbf{X}}_1'\hat{\mathbf{e}}_1\dot{x}_{1,T_1^*}' - \tilde{e}_{1,T_1^*}\dot{\mathbf{X}}_1'\hat{\mathbf{e}}_1\mathbf{A}'(\dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1 + \kappa\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}') \\
&+ \kappa\tilde{e}_{1,T_1^*}\dot{x}_{1,T_1^*}\hat{\mathbf{e}}_1'\dot{\mathbf{X}}_1 - \tilde{e}_{1,T_1^*}(\dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1 + \kappa\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}')\mathbf{A}\hat{\mathbf{e}}_1'\dot{\mathbf{X}}_1 \\
&+ \tilde{e}_{1,T_1^*}^2\big[\kappa^2\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}' - \kappa\dot{x}_{1,T_1^*}\mathbf{A}'(\dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1 + \kappa\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}') - \kappa(\dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1 + \kappa\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}')\mathbf{A}\dot{x}_{1,T_1^*}' \\
&+ (\dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1 + \kappa\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}')\mathbf{A}\mathbf{A}'(\dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1 + \kappa\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}')\big].
\end{aligned}
$$

The last equation can be arranged as follows:

$$
\begin{aligned}
&\dot{\mathbf{X}}_1'\hat{\mathbf{e}}_1\hat{\mathbf{e}}_1'\dot{\mathbf{X}}_1 - \tilde{e}_{1,T_1^*}\dot{\mathbf{X}}_1'\hat{\mathbf{e}}_1\mathbf{A}'\dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1 - \tilde{e}_{1,T_1^*}\dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1\mathbf{A}\hat{\mathbf{e}}_1'\dot{\mathbf{X}}_1 + \tilde{e}_{1,T_1^*}^2\dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1\mathbf{A}\mathbf{A}'\dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1 && \text{(A.14)}\\
&+\kappa\tilde{e}_{1,T_1^*}\dot{\mathbf{X}}_1'\hat{\mathbf{e}}_1\dot{x}_{1,T_1^*}' - \kappa\tilde{e}_{1,T_1^*}\dot{\mathbf{X}}_1'\hat{\mathbf{e}}_1\mathbf{A}'\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}' && \text{(A.15)}\\
&+\kappa\tilde{e}_{1,T_1^*}\dot{x}_{1,T_1^*}\hat{\mathbf{e}}_1'\dot{\mathbf{X}}_1 - \kappa\tilde{e}_{1,T_1^*}\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}'\mathbf{A}\hat{\mathbf{e}}_1'\dot{\mathbf{X}}_1 && \text{(A.16)}\\
&+\kappa^2\tilde{e}_{1,T_1^*}^2\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}'\dot{x}_{1,T_1^*}' - \kappa^2\tilde{e}_{1,T_1^*}^2\dot{x}_{1,T_1^*}\mathbf{A}'\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}' && \text{(A.17)}\\
&-\kappa\tilde{e}_{1,T_1^*}^2\dot{x}_{1,T_1^*}\mathbf{A}'\dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1 + \kappa\tilde{e}_{1,T_1^*}^2\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}'\mathbf{A}\mathbf{A}'\dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1 && \text{(A.18)}\\
&-\kappa\tilde{e}_{1,T_1^*}^2\dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1\mathbf{A}\dot{x}_{1,T_1^*}' + \kappa\tilde{e}_{1,T_1^*}^2\dot{\mathbf{X}}_1'\dot{\mathbf{X}}_1\mathbf{A}\mathbf{A}'\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}' && \text{(A.19)}\\
&-\kappa^2\tilde{e}_{1,T_1^*}^2\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}'\dot{x}_{1,T_1^*}'\mathbf{A}\dot{x}_{1,T_1^*}' + \kappa^2\tilde{e}_{1,T_1^*}^2\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}'\dot{x}_{1,T_1^*}'\mathbf{A}\mathbf{A}'\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}'. && \text{(A.20)}
\end{aligned}
$$

(A.15) can be rewritten as

$$
\begin{aligned}
&\kappa\tilde{e}_{1,T_1^*}\dot{\mathbf{X}}_1'\hat{\mathbf{e}}_1\dot{x}_{1,T_1^*}' - \kappa\tilde{e}_{1,T_1^*}\dot{\mathbf{X}}_1'\hat{\mathbf{e}}_1\mathbf{A}'\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}' \\
=&\kappa\tilde{e}_{1,T_1^*}\dot{\mathbf{X}}_1'\hat{\mathbf{e}}_1\left(1 - \mathbf{A}'\dot{x}_{1,T_1^*}\right)\dot{x}_{1,T_1^*}' \\
=&\kappa\tilde{e}_{1,T_1^*}\dot{\mathbf{X}}_1'\hat{\mathbf{e}}_1\left(1 - \frac{\kappa h_{1,T_1^*}}{1+\kappa h_{1,T_1^*}}\right)\dot{x}_{1,T_1^*}' \\
=&s_{1,T_1^*}\tilde{e}_{1,T_1^*}\dot{\mathbf{X}}_1'\hat{\mathbf{e}}_1\dot{x}_{1,T_1^*}',
\end{aligned}
$$

and (A.16) is the transpose of (A.15).

(A.17) can be rewritten as

$$
\begin{aligned}
&\kappa^2\tilde{e}^2_{1,T_1^*}\dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*} - \kappa^2\tilde{e}^2_{1,T_1^*}\dot{x}_{1,T_1^*}\mathbf{A}'\dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*}\\
=&\kappa^2\tilde{e}^2_{1,T_1^*}\dot{x}_{1,T_1^*}\left(1-\mathbf{A}'\dot{x}_{1,T_1^*}\right)\dot{x}'_{1,T_1^*}\\
=&\kappa s_{1,T_1^*}\tilde{e}^2_{1,T_1^*}\dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*}
\end{aligned}
$$

(A.18) can be rewritten as

$$
\begin{aligned}
&-\kappa\tilde{e}^2_{1,T_1^*}\dot{x}_{1,T_1^*}\mathbf{A}'\dot{\mathbf{X}}'_1\dot{\mathbf{X}}_1 + \kappa\tilde{e}^2_{1,T_1^*}\dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*}\mathbf{A}\mathbf{A}'\dot{\mathbf{X}}'_1\dot{\mathbf{X}}_1\\
=&-\kappa\tilde{e}^2_{1,T_1^*}\dot{x}_{1,T_1^*}\left(1-\dot{x}'_{1,T_1^*}\mathbf{A}\right)\mathbf{A}'\dot{\mathbf{X}}'_1\dot{\mathbf{X}}_1\\
=&-s^2_{1,T_1^*}\tilde{e}^2_{1,T_1^*}\dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}'_1\dot{\mathbf{X}}_1,
\end{aligned}
$$

and (A.19) is the transpose of (A.18).

Finally, (A.20) can be rewritten as

$$
\begin{aligned}
&-\kappa^2\tilde{e}^2_{1,T_1^*}\dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*}\mathbf{A}\dot{x}'_{1,T_1^*} + \kappa^2\tilde{e}^2_{1,T_1^*}\dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*}\mathbf{A}\mathbf{A}'\dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*}\\
=&-\kappa^2\tilde{e}^2_{1,T_1^*}\dot{x}_{1,T_1^*}\dot{x}_{1,T_1^*}\mathbf{A}\left(1-\mathbf{A}'\dot{x}_{1,T_1^*}\right)\dot{x}'_{1,T_1^*}\\
=&-\kappa s^2_{1,T_1^*}\tilde{e}^2_{1,T_1^*}h_{1,T_1^*}\dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*}
\end{aligned}
$$

Combining those results, (A.15)-(A.20) become

$$
\begin{aligned}
&s_{1,T_1^*}\tilde{e}_{1,T_1^*}\dot{\mathbf{X}}'_1\hat{\mathbf{e}}_1\dot{x}'_{1,T_1^*} + s_{1,T_1^*}\tilde{e}_{1,T_1^*}\dot{x}_{1,T_1^*}\hat{\mathbf{e}}'_1\dot{\mathbf{X}}_1\\
&+\kappa s_{1,T_1^*}\tilde{e}^2_{1,T_1^*}\dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*}\\
&-s^2_{1,T_1^*}\tilde{e}^2_{1,T_1^*}\dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}'_1\dot{\mathbf{X}}_1 - s^2_{1,T_1^*}\tilde{e}^2_{1,T_1^*}\dot{\mathbf{X}}'_1\dot{\mathbf{X}}_1(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*}\\
&-\kappa s^2_{1,T_1^*}\tilde{e}^2_{1,T_1^*}h_{1,T_1^*}\dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*}\\
=&s_{1,T_1^*}\tilde{e}_{1,T_1^*}(\dot{\mathbf{X}}'_1\hat{\mathbf{e}}_1\dot{x}'_{1,T_1^*} + \dot{x}_{1,T_1^*}\hat{\mathbf{e}}'_1\dot{\mathbf{X}}_1)\\
&+\kappa s_{1,T_1^*}\tilde{e}^2_{1,T_1^*}\dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*} - \kappa s^2_{1,T_1^*}\tilde{e}^2_{1,T_1^*}h_{1,T_1^*}\dot{x}_{1,T_1^*}\dot{x}'_{1,T_1^*}
\end{aligned}
$$

$$
\begin{aligned}
&- s_{1,T_1^*}^2 \tilde{e}_{1,T_1^*}^2 \dot{x}_{1,T_1^*} \dot{x}_{1,T_1^*}' (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1} \dot{\mathbf{X}}_1' \dot{\mathbf{X}}_1 - s_{1,T_1^*}^2 \tilde{e}_{1,T_1^*}^2 \dot{\mathbf{X}}_1' \dot{\mathbf{X}}_1 (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1} \dot{x}_{1,T_1^*} \dot{x}_{1,T_1^*}' \\
=& s_{1,T_1^*} \tilde{e}_{1,T_1^*} (\dot{\mathbf{X}}_1' \hat{\mathbf{e}}_1 \dot{x}_{1,T_1^*}' + \dot{x}_{1,T_1^*} \hat{\mathbf{e}}_1' \dot{\mathbf{X}}_1) \\
&+ s_{1,T_1^*}^2 \tilde{e}_{1,T_1^*}^2 \dot{x}_{1,T_1^*} \dot{x}_{1,T_1^*}' - s_{1,T_1^*}^2 \tilde{e}_{1,T_1^*}^2 \dot{x}_{1,T_1^*} \dot{x}_{1,T_1^*}' (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1} \dot{\mathbf{X}}_1' \dot{\mathbf{X}}_1 - s_{1,T_1^*}^2 \tilde{e}_{1,T_1^*}^2 \dot{\mathbf{X}}_1' \dot{\mathbf{X}}_1 (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1} \dot{x}_{1,T_1^*} \dot{x}_{1,T_1^*}' \\
=& s_{1,T_1^*} \tilde{e}_{1,T_1^*} (\dot{\mathbf{X}}_1' \hat{\mathbf{e}}_1 \dot{x}_{1,T_1^*}' + \dot{x}_{1,T_1^*} \hat{\mathbf{e}}_1' \dot{\mathbf{X}}_1) \\
&+ s_{1,T_1^*}^2 \tilde{e}_{1,T_1^*}^2 \left(\dot{x}_{1,T_1^*} \dot{x}_{1,T_1^*}' - \dot{x}_{1,T_1^*} \dot{x}_{1,T_1^*}' (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1} \dot{\mathbf{X}}_1' \dot{\mathbf{X}}_1 - \dot{\mathbf{X}}_1' \dot{\mathbf{X}}_1 (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1} \dot{x}_{1,T_1^*} \dot{x}_{1,T_1^*}' \right)
\end{aligned}
$$

$\sum_{i=2}^N \dot{\mathbf{X}}_i' \hat{\mathbf{e}}_i^* \hat{\mathbf{e}}_i^{*\prime} \dot{\mathbf{X}}_i$ can be written as follows:

$$
\begin{aligned}
&\sum_{i=2}^N \dot{\mathbf{X}}_i' \left(\hat{\mathbf{e}}_i - \dot{\mathbf{X}}_i \mathbf{A} \tilde{e}_{1,T_1^*} \right) \left(\hat{\mathbf{e}}_i - \dot{\mathbf{X}}_i \mathbf{A} \tilde{e}_{1,T_1^*} \right)' \dot{\mathbf{X}}_i \\
=& \sum_{i=2}^N \dot{\mathbf{X}}_i' \left(\hat{\mathbf{e}}_i \hat{\mathbf{e}}_i' - \hat{\mathbf{e}}_i \mathbf{A}' \dot{\mathbf{X}}_i' \tilde{e}_{1,T_1^*} - \dot{\mathbf{X}}_i \mathbf{A} \hat{\mathbf{e}}_i' \tilde{e}_{1,T_1^*} + \dot{\mathbf{X}}_i \mathbf{A}\mathbf{A}' \dot{\mathbf{X}}_i' \tilde{e}_{1,T_1^*}^2 \right) \dot{\mathbf{X}}_i \\
=& \sum_{i=2}^N \dot{\mathbf{X}}_i' \hat{\mathbf{e}}_i \hat{\mathbf{e}}_i' \dot{\mathbf{X}}_i - \tilde{e}_{1,T_1^*} \sum_{i=2}^N \dot{\mathbf{X}}_i' \hat{\mathbf{e}}_i \mathbf{A}' \dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i - \tilde{e}_{1,T_1^*} \sum_{i=2}^N \dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i \mathbf{A} \hat{\mathbf{e}}_i' \dot{\mathbf{X}}_i + \tilde{e}_{1,T_1^*}^2 \sum_{i=2}^N \dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i \mathbf{A}\mathbf{A}' \dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i
\end{aligned}
$$

Using the results derived at the above steps, we can rewrite $\sum_{i=1}^N \dot{\mathbf{X}}_i^{*\prime} \hat{\mathbf{e}}_i^* \hat{\mathbf{e}}_i^{*\prime} \dot{\mathbf{X}}_i^*$ as follows:

$$
\begin{aligned}
&\dot{\mathbf{X}}_1' \hat{\mathbf{e}}_1 \hat{\mathbf{e}}_1' \dot{\mathbf{X}}_1 - \tilde{e}_{1,T_1^*} \dot{\mathbf{X}}_1' \hat{\mathbf{e}}_1 \mathbf{A}' \dot{\mathbf{X}}_1' \dot{\mathbf{X}}_1 - \tilde{e}_{1,T_1^*} \dot{\mathbf{X}}_1' \dot{\mathbf{X}}_1 \mathbf{A} \hat{\mathbf{e}}_1' \dot{\mathbf{X}}_1 + \tilde{e}_{1,T_1^*}^2 \dot{\mathbf{X}}_1' \dot{\mathbf{X}}_1 \mathbf{A}\mathbf{A}' \dot{\mathbf{X}}_1' \dot{\mathbf{X}}_1 \\
&+ s_{1,T_1^*} \tilde{e}_{1,T_1^*} (\dot{\mathbf{X}}_1' \hat{\mathbf{e}}_1 \dot{x}_{1,T_1^*}' + \dot{x}_{1,T_1^*} \hat{\mathbf{e}}_1' \dot{\mathbf{X}}_1) \\
&+ s_{1,T_1^*}^2 \tilde{e}_{1,T_1^*}^2 \left(\dot{x}_{1,T_1^*} \dot{x}_{1,T_1^*}' - \dot{x}_{1,T_1^*} \dot{x}_{1,T_1^*}' (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1} \dot{\mathbf{X}}_1' \dot{\mathbf{X}}_1 - \dot{\mathbf{X}}_1' \dot{\mathbf{X}}_1 (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1} \dot{x}_{1,T_1^*} \dot{x}_{1,T_1^*}' \right) \\
&+ \sum_{i=2}^N \dot{\mathbf{X}}_i' \hat{\mathbf{e}}_i \hat{\mathbf{e}}_i' \dot{\mathbf{X}}_i - \tilde{e}_{1,T_1^*} \sum_{i=2}^N \dot{\mathbf{X}}_i' \hat{\mathbf{e}}_i \mathbf{A}' \dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i - \tilde{e}_{1,T_1^*} \sum_{i=2}^N \dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i \mathbf{A} \hat{\mathbf{e}}_i' \dot{\mathbf{X}}_i + \tilde{e}_{1,T_1^*}^2 \sum_{i=2}^N \dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i \mathbf{A}\mathbf{A}' \dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i \\
=& \sum_{i=1}^N \dot{\mathbf{X}}_i' \hat{\mathbf{e}}_i \hat{\mathbf{e}}_i' \dot{\mathbf{X}}_i - \tilde{e}_{1,T_1^*} \sum_{i=1}^N \dot{\mathbf{X}}_i' \hat{\mathbf{e}}_i \mathbf{A}' \dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i - \tilde{e}_{1,T_1^*} \sum_{i=1}^N \dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i \mathbf{A} \hat{\mathbf{e}}_i' \dot{\mathbf{X}}_i + \tilde{e}_{1,T_1^*}^2 \sum_{i=1}^N \dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i \mathbf{A}\mathbf{A}' \dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i \\
&+ s_{1,T_1^*} \tilde{e}_{1,T_1^*} (\dot{\mathbf{X}}_1' \hat{\mathbf{e}}_1 \dot{x}_{1,T_1^*}' + \dot{x}_{1,T_1^*} \hat{\mathbf{e}}_1' \dot{\mathbf{X}}_1) \\
&+ s_{1,T_1^*}^2 \tilde{e}_{1,T_1^*}^2 \left(\dot{x}_{1,T_1^*} \dot{x}_{1,T_1^*}' - \dot{x}_{1,T_1^*} \dot{x}_{1,T_1^*}' (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1} \dot{\mathbf{X}}_1' \dot{\mathbf{X}}_1 - \dot{\mathbf{X}}_1' \dot{\mathbf{X}}_1 (\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1} \dot{x}_{1,T_1^*} \dot{x}_{1,T_1^*}' \right)
\end{aligned}
$$

We can rewrite the matrices in the last equation above using the vectorization property, as done in deriving cluster-robust when unit data arrive. Then, the matrices can be rewritten as

follows:

$$
\begin{aligned}
\Omega^{(1)}_{1,T_1^*} :=& \tilde{e}_{1,T_1^*} \sum_{i=1}^{N} \dot{\mathbf{X}}_i' \hat{\mathbf{e}}_i \mathbf{A}' \dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i \\
=& \tilde{e}_{1,T_1^*} \sum_{i=1}^{N} \dot{\mathbf{X}}_i' (\dot{\mathbf{Y}}_i - \dot{\mathbf{X}}_i \hat{\beta}_{fe}) \operatorname{vec}(\dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i)' (\mathbf{I}_k \otimes \mathbf{A}) \\
=& \tilde{e}_{1,T_1^*} \left(\sum_{i=1}^{N} \dot{\mathbf{X}}_i' \dot{\mathbf{Y}}_i \operatorname{vec}(\dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i)' - \sum_{i=1}^{N} \dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i \hat{\beta}_{fe} \operatorname{vec}(\dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i)' \right) (\mathbf{I}_k \otimes \mathbf{A}) \\
=& \tilde{e}_{1,T_1^*} \left\{ \sum_{i=1}^{N} \dot{\mathbf{X}}_i' \dot{\mathbf{Y}}_i \operatorname{vec}(\dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i)' - (\hat{\beta}_{fe}' \otimes \mathbf{I}_k) \left(\sum_{i=1}^{N} \operatorname{vec}(\dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i) \operatorname{vec}(\dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i)' \right) \right\} (\mathbf{I}_k \otimes \mathbf{A}) \\
=& \tilde{e}_{1,T_1^*} \left\{ \sum_{i=1}^{N} \dot{\mathbf{X}}_i' \dot{\mathbf{Y}}_i \operatorname{vec}(\dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i)' - (\hat{\beta}_{fe}' \otimes \mathbf{I}_k) \left(\sum_{i=1}^{N} \operatorname{vec}(\dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i) \operatorname{vec}(\dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i)' \right) \right\} \left(\mathbf{I}_k \otimes s_{1,T_1^*} (\dot{\mathbf{X}}' \dot{\mathbf{X}})^{-1} \dot{x}_{1,T_1^*} \right) \\
\Omega^{(2)}_{1,T_1^*} :=& \tilde{e}^2_{1,T_1^*} \sum_{i=1}^{N} \dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i \mathbf{A} \mathbf{A}' \dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i \\
=& \tilde{e}^2_{1,T_1^*} (s_{1,T_1^*} \dot{x}_{1,T_1^*}' (\dot{\mathbf{X}}' \dot{\mathbf{X}})^{-1} \otimes \mathbf{I}_k) \left(\sum_{i=1}^{N} \operatorname{vec}(\dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i) \operatorname{vec}(\dot{\mathbf{X}}_i' \dot{\mathbf{X}}_i)' \right) (\mathbf{I}_k \otimes s_{1,T_1^*} (\dot{\mathbf{X}}' \dot{\mathbf{X}})^{-1} \dot{x}_{1,T_1^*}).
\end{aligned}
$$

Define

$$
\begin{aligned}
\Omega^{(3)}_{1,T_1^*} =& s_{1,T_1^*} \tilde{e}_{1,T_1^*} (\dot{\mathbf{X}}_1' \hat{\mathbf{e}}_1 \dot{x}_{1,T_1^*}' + \dot{x}_{1,T_1^*} \hat{\mathbf{e}}_1' \dot{\mathbf{X}}_1), \\
\Omega^{(4)}_{1,T_1^*} =& s^2_{1,T_1^*} \tilde{e}^2_{1,T_1^*} \left(\dot{x}_{1,T_1^*} \dot{x}_{1,T_1^*}' - \dot{x}_{1,T_1^*} \dot{x}_{1,T_1^*}' (\dot{\mathbf{X}}' \dot{\mathbf{X}})^{-1} \dot{\mathbf{X}}_1' \dot{\mathbf{X}}_1 - \dot{\mathbf{X}}_1' \dot{\mathbf{X}}_1 (\dot{\mathbf{X}}' \dot{\mathbf{X}})^{-1} \dot{x}_{1,T_1^*} \dot{x}_{1,T_1^*}' \right).
\end{aligned}
$$

Then, the updated cluster-robust is

$$
\begin{aligned}
\hat{V}^{cr*}_{2,fe} =& \mathbf{U}_{1,T_1^*} (\dot{\mathbf{X}}' \dot{\mathbf{X}})^{-1} \left(\sum_{i=1}^{N} \dot{\mathbf{X}}_i' \hat{\mathbf{e}}_i \hat{\mathbf{e}}_i' \dot{\mathbf{X}}_i - \Omega^{(1)}_{1,T_1^*} - \Omega^{(1)'}_{1,T_1^*} + \Omega^{(2)}_{1,T_1^*} + \Omega^{(3)}_{1,T_1^*} + \Omega^{(4)}_{1,T_1^*} \right) (\dot{\mathbf{X}}' \dot{\mathbf{X}})^{-1} \mathbf{U}_{1,T_1^*}' \\
=& \mathbf{U}_{1,T_1^*} \left[\hat{V}^{cr}_{fe} + (\dot{\mathbf{X}}' \dot{\mathbf{X}})^{-1} \left(-\Omega^{(1)}_{1,T_1^*} - \Omega^{(1)'}_{1,T_1^*} + \Omega^{(2)}_{1,T_1^*} + \Omega^{(3)}_{1,T_1^*} + \Omega^{(4)}_{1,T_1^*} \right) (\dot{\mathbf{X}}' \dot{\mathbf{X}})^{-1} \right] \mathbf{U}_{1,T_1^*}',
\end{aligned}
$$

where $\mathbf{U}_{1,T_1^*} := \mathbf{I}_k - s_{1,T_1^*} (\dot{\mathbf{X}}' \dot{\mathbf{X}})^{-1} \dot{x}_{1,T_1^*} \dot{x}_{1,T_1^*}'$

A.8 Proposition 7

We prove (4.3), (4.4), and (4.5) in turn. The incoming observation arrives at calendar time $t_{\text{new}} = T + 1$, so a new time dummy column D^{T+1} is appended and the regressor dimension increases from p to $p + 1$. For each pre-update observation (i, t), we have $D_{it}^{T+1} = 0$, so we embed the raw regressor vector into $\mathbb{R}^{p+1}$ by

$$\tilde{Z}_{it} := \begin{bmatrix} Z_{it} \\ 0 \end{bmatrix} \in \mathbb{R}^{p+1}, \quad \bar{\tilde{Z}}_i := \begin{bmatrix} \bar{Z}_i \\ 0 \end{bmatrix} \in \mathbb{R}^{p+1}.$$

The incoming observation contributes the expanded regressor vector $\left[Z'_{1,t_{\text{new}}},\, D_{1,t_{\text{new}}}^{T+1}\right]' \in \mathbb{R}^{p+1}$ with $D_{1,t_{\text{new}}}^{T+1} = 1$. Define the incoming within-deviation in $\mathbb{R}^{p+1}$ by

$$g := \begin{bmatrix} Z_{1,t_{\text{new}}} \\ D_{1,t_{\text{new}}}^{T+1} \end{bmatrix} - \begin{bmatrix} \bar{Z}_1 \\ 0 \end{bmatrix} = \begin{bmatrix} g_1 \\ g_2 \end{bmatrix}, \quad g_1 \in \mathbb{R}^p,\ g_2 \in \mathbb{R},$$

so that $g_1 = Z_{1,t_{\text{new}}} - \bar{Z}_1$ and $g_2 = D_{1,t_{\text{new}}}^{T+1} = 1$ in our setup. The post-update mean in $\mathbb{R}^{p+1}$ is

$$\bar{\tilde{Z}}_1^* = \frac{1}{T_1 + 1}\left(\sum_{t \in S_1} \tilde{Z}_{1t} + \begin{bmatrix} Z_{1,t_{\text{new}}} \\ D_{1,t_{\text{new}}}^{T+1} \end{bmatrix}\right) = \bar{\tilde{Z}}_1 + \frac{1}{T_1 + 1} g.$$

Therefore, for any old observation $(1, t)$ with $t \in S_1$,

$$\dot{Z}_{1t}^* := \tilde{Z}_{1t} - \bar{\tilde{Z}}_1^* = \begin{bmatrix} \dot{Z}_{1t} \\ 0 \end{bmatrix} - \frac{1}{T_1 + 1} g,$$

where $\dot{Z}_{1t} := Z_{1t} - \bar{Z}_1$ is the pre-update within-deviation in $\mathbb{R}^p$. For the new observation,

$$\dot{Z}_{1,t_{\text{new}}}^* := \begin{bmatrix} Z_{1,t_{\text{new}}} \\ D_{1,t_{\text{new}}}^{T+1} \end{bmatrix} - \bar{\tilde{Z}}_1^* = g - \frac{1}{T_1 + 1} g = \kappa g, \quad \kappa := \frac{T_1}{T_1 + 1}.$$

Finally, for any $i \neq 1$, the individual mean is unchanged, hence for all pre-update observations $t \in S_i$,

$$\dot{Z}^*_{it} = \begin{bmatrix} \dot{Z}_{it} \\ 0 \end{bmatrix}.$$

By definition,

$$\dot{\mathbf{Z}}^{*\prime}\dot{\mathbf{Z}}^* = \sum_{i=1}^{N} \sum_{t \in S^*_i} \dot{Z}^*_{it} \dot{Z}^{*\prime}_{it}.$$

We decompose the sum into (i) pre-update observations of individuals $i \neq 1$, (ii) pre-update observations of individual 1, and (iii) the new observation $(1, t_{\text{new}})$.

For (i), since the appended coordinate is zero,

$$\sum_{i \neq 1} \sum_{t \in S_i} \begin{bmatrix} \dot{Z}_{it} \\ 0 \end{bmatrix} \begin{bmatrix} \dot{Z}_{it} \\ 0 \end{bmatrix}' = \begin{bmatrix} \sum_{i \neq 1} \sum_{t \in S_i} \dot{Z}_{it} \dot{Z}'_{it} & 0 \\ 0 & 0 \end{bmatrix}.$$

For (ii), writing $a_t := [\dot{Z}'_{1t}, 0]'$ and $\alpha := 1/(T_1 + 1)$,

$$\sum_{t \in S_1} (a_t - \alpha g)(a_t - \alpha g)' = \sum_{t \in S_1} a_t a'_t - \alpha \left(\sum_{t \in S_1} a_t \right) g' - \alpha g \left(\sum_{t \in S_1} a_t \right)' + T_1 \alpha^2 g g'.$$

The cross terms vanish because $\sum_{t \in S_1} \dot{Z}_{1t} = 0$ implies $\sum_{t \in S_1} a_t = 0$. Hence, the contribution of (ii) equals

$$\sum_{t \in S_1} \begin{bmatrix} \dot{Z}_{1t} \\ 0 \end{bmatrix} \begin{bmatrix} \dot{Z}_{1t} \\ 0 \end{bmatrix}' + \frac{T_1}{(T_1 + 1)^2} g g'.$$

For (iii), the new observation contributes $(\kappa g)(\kappa g)' = \kappa^2 g g'$. Combining (i)–(iii) and noting

that $\sum_{i\neq 1}\sum_{t\in S_i}\dot{Z}_{it}\dot{Z}_{it}' + \sum_{t\in S_1}\dot{Z}_{1t}\dot{Z}_{1t}' = \dot{\mathbf{Z}}'\dot{\mathbf{Z}}$, we obtain

$$\dot{\mathbf{Z}}^{*\prime}\dot{\mathbf{Z}}^{*} = \begin{bmatrix} \dot{\mathbf{Z}}'\dot{\mathbf{Z}} & 0 \\ 0 & 0 \end{bmatrix} + \left(\frac{T_1}{(T_1+1)^2} + \kappa^2\right) gg'.$$

Since $\kappa = T_1/(T_1+1)$, we have $\frac{T_1}{(T_1+1)^2} + \kappa^2 = \kappa$, which proves (4.3).

To obtain the inverse, write (4.3) in block form using $g = (g_1', g_2)'$:

$$\dot{\mathbf{Z}}^{*\prime}\dot{\mathbf{Z}}^{*} = \begin{bmatrix} \dot{\mathbf{Z}}'\dot{\mathbf{Z}} + \kappa g_1 g_1' & \kappa g_1 g_2 \\ \kappa g_2 g_1' & \kappa g_2^2 \end{bmatrix}.$$

Let $A := (\dot{\mathbf{Z}}'\dot{\mathbf{Z}})^{-1}$, $q := Ag_1$, and $h := g_1'Ag_1 = g_1'q$. A straightforward matrix algebra verifies that the matrix

$$\begin{bmatrix} A & -\dfrac{q}{g_2} \\ -\dfrac{q'}{g_2} & \dfrac{1+\kappa h}{\kappa g_2^2} \end{bmatrix}$$

is the inverse of $\dot{\mathbf{Z}}^{*\prime}\dot{\mathbf{Z}}^{*}$, which establishes (4.4).

We next characterize $\dot{\mathbf{Z}}^{*\prime}\dot{\mathbf{Y}}^{*}$. Using the obtained results and the fact that within-transformed outcomes sum to zero within an individual, $\sum_{t\in S_1}(Y_{1t} - \bar{Y}_1) = 0$, the cross terms vanish and we obtain

$$\dot{\mathbf{Z}}^{*\prime}\dot{\mathbf{Y}}^{*} = \begin{bmatrix} \dot{\mathbf{Z}}'\dot{\mathbf{Y}} \\ 0 \end{bmatrix} + \kappa g \dot{y}_{1,t_{\text{new}}}. \tag{A.21}$$

Let $\hat{\theta} := (\dot{\mathbf{Z}}'\dot{\mathbf{Z}})^{-1}\dot{\mathbf{Z}}'\dot{\mathbf{Y}}$. Multiplying (4.4) by (A.21) yields

$$\hat{\theta}^{*} = \begin{bmatrix} \hat{\theta} \\ 0 \end{bmatrix} + \kappa \begin{bmatrix} Ag_1 - \dfrac{Ag_1}{g_2}\, g_2 \\ -\dfrac{g_1'\hat{\theta}}{g_2} + \dfrac{1+\kappa h}{\kappa g_2^2}\, g_2 \end{bmatrix} \dot{y}_{1,t_{\text{new}}},$$

so the upper p entries equal $\hat{\theta}$. For the last entry, using $h = g_1' A g_1$ and simplifying, we obtain

$$\hat{v}_{T+1} = \frac{\dot{y}_{1,t_{\text{new}}} - g_1'\hat{\beta}}{g_2},$$

which proves (4.5).

A.9 Proposition 8

Recall that the pre-update residual vector is

$$\hat{\boldsymbol{\epsilon}} := \dot{\mathbf{Y}} - \dot{\mathbf{Z}}\hat{\theta}, \quad \hat{\sigma}^2 = (n - N - p)^{-1}\hat{\boldsymbol{\epsilon}}'\hat{\boldsymbol{\epsilon}}, \quad \hat{V}_{fe}^0 = \hat{\sigma}^2(\dot{\mathbf{Z}}'\dot{\mathbf{Z}})^{-1}.$$

After the update, define

$$\hat{\boldsymbol{\epsilon}}^* := \dot{\mathbf{Y}}^* - \dot{\mathbf{Z}}^*\hat{\theta}^*, \quad \hat{\sigma}^{2*} = (n^* - N - p^*)^{-1}\hat{\boldsymbol{\epsilon}}^{*\prime}\hat{\boldsymbol{\epsilon}}^*, \quad \hat{V}_{fe}^{0*} = \hat{\sigma}^{2*}(\dot{\mathbf{Z}}^{*\prime}\dot{\mathbf{Z}}^*)^{-1}.$$

One new observation is added and one new time dummy column is appended, so $n^* = n+1$ and $p^* = p+1$. Then, the degrees of freedom are unchanged:

$$n^* - N - p^* = (n+1) - N - (p+1) = n - N - p.$$

Therefore, it suffices to show $\hat{\boldsymbol{\epsilon}}^{*\prime}\hat{\boldsymbol{\epsilon}}^* = \hat{\boldsymbol{\epsilon}}'\hat{\boldsymbol{\epsilon}}$. By Proposition 7, the updated coefficient vector satisfies

$$\hat{\theta}^* = \begin{bmatrix} \hat{\theta} \\ \hat{v}_{T+1} \end{bmatrix}, \quad \hat{v}_{T+1} = \frac{\dot{y}_{1,t_{\text{new}}} - g_1'\hat{\theta}}{g_2},$$

where $\dot{y}_{1,t_{\text{new}}} := Y_{1,t_{\text{new}}} - \bar{Y}_1$ and $g = (g_1', g_2)'$ is defined in (4.2). Consequently,

$$g'\hat{\theta}^* = g_1'\hat{\theta} + g_2\hat{v}_{T+1} = \dot{y}_{1,t_{\text{new}}}. \tag{A.22}$$

We now compare residuals row-by-row. From the proof of Proposition 7, the post-update within-transformed regressor and outcome for the incoming observation satisfy

$$\dot{Z}^*_{1,t_{\text{new}}} = \kappa g, \quad \dot{Y}^*_{1,t_{\text{new}}} = \kappa \dot{y}_{1,t_{\text{new}}}, \quad \kappa := \frac{T_1}{T_1+1}.$$

Using (A.22), the post-update residual is exactly zero:

$$\hat{\epsilon}^*_{1,t_{\text{new}}} = \dot{Y}^*_{1,t_{\text{new}}} - (\dot{Z}^*_{1,t_{\text{new}}})'\hat{\theta}^* = \kappa\left(\dot{y}_{1,t_{\text{new}}} - g'\hat{\theta}^*\right) = 0.$$

For any pre-update observation (i,t), the appended time dummy coordinate equals 0. Moreover, if $i \neq 1$, the individual mean is unchanged, while if $i = 1$, both $\dot{Z}^*_{1t}$ and $\dot{Y}^*_{1t}$ are shifted by the same mean-update term. In both cases, one obtains

$$\hat{\epsilon}^*_{it} = \hat{\epsilon}_{it}.$$

(For $i = 1$, this follows by substituting $\dot{Z}^*_{1t} = \left[\dot{Z}'_{1t},\, 0\right]' - \frac{1}{T_1+1} g$ and $\dot{Y}^*_{1t} = \dot{Y}_{1t} - \frac{1}{T_1+1}\dot{y}_{1,t_{\text{new}}}$ and using (A.22).)

Combining the above results, all pre-update residuals are unchanged and the new residual equals 0. Therefore,

$$\hat{\boldsymbol{\epsilon}}^{*\prime}\hat{\boldsymbol{\epsilon}}^* = \hat{\boldsymbol{\epsilon}}'\hat{\boldsymbol{\epsilon}}.$$

Since $n^* - N - p^* = n - N - p$, it follows that $\hat{\sigma}^{2*} = \hat{\sigma}^2$, and hence

$$\hat{V}^{0*}_{fe} = \hat{\sigma}^{2*}(\dot{\mathbf{Z}}^{*\prime}\dot{\mathbf{Z}}^*)^{-1} = \hat{\sigma}^2(\dot{\mathbf{Z}}^{*\prime}\dot{\mathbf{Z}}^*)^{-1},$$

which proves (4.6). Finally, substituting (4.4) yields (4.7).

A.10 Proposition 9

Recall that the pre-update residuals are $\hat{\mathbf{e}}_i := \dot{\mathbf{Y}}_i - \dot{\mathbf{Z}}_i\hat{\theta}$ and the post-update residuals are $\hat{\mathbf{e}}_i^* := \dot{\mathbf{Y}}_i^* - \dot{\mathbf{Z}}_i^*\hat{\theta}^*$. Define $r_i := \dot{\mathbf{Z}}_i'\hat{\mathbf{e}}_i \in \mathbb{R}^p$ and $r_i^* := \dot{\mathbf{Z}}_i^{*\prime}\hat{\mathbf{e}}_i^* \in \mathbb{R}^{p+1}$, so that $\mathbf{M} = \sum_{i=1}^N r_i r_i'$ and $\mathbf{M}^* = \sum_{i=1}^N r_i^* r_i^{*\prime}$. The new time dummy column is appended and is identically zero for all pre-update observations. Hence, for any $i \neq 1$ and $t \in S_i$, $\dot{Z}_{it}^* = [\dot{Z}_{it}', 0]'$ and $\hat{e}_{it}^* = \hat{e}_{it}$, so $r_i^* = \tilde{r}_i$ for all $i = 2, \cdots, N$, where $\tilde{r}_i := [r_i', 0]'$. Therefore,

$$\mathbf{M}^* = \sum_{i=2}^N \tilde{r}_i\tilde{r}_i' + r_1^* r_1^{*\prime} = \widetilde{\mathbf{M}} - \tilde{r}_1\tilde{r}_1' + r_1^* r_1^{*\prime}. \tag{A.23}$$

It remains to give an exact updating formula for r_1^*. Write the stored summaries as

$$s_1 := \dot{\mathbf{Z}}_1'\dot{\mathbf{Y}}_1, \quad S_1 := \dot{\mathbf{Z}}_1'\dot{\mathbf{Z}}_1, \quad \tilde{r}_1 = \begin{bmatrix} s_1 - S_1\hat{\theta} \\ 0 \end{bmatrix}.$$

Under the new-time update for individual 1, the within-transformed summaries satisfy

$$\dot{\mathbf{Z}}_1^{*\prime}\dot{\mathbf{Z}}_1^* = \begin{bmatrix} S_1 & 0 \\ 0 & 0 \end{bmatrix} + \kappa g g' = \tilde{S}_1 + \kappa g g', \quad \dot{\mathbf{Z}}_1^{*\prime}\dot{\mathbf{Y}}_1^* = \begin{bmatrix} s_1 \\ 0 \end{bmatrix} + \kappa g \dot{y}_{1,t_{\text{new}}} = \tilde{s}_1 + \kappa g \dot{y}_{1,t_{\text{new}}},$$

where $\tilde{S}_1 = \begin{bmatrix} S_1 & 0 \\ 0 & 0 \end{bmatrix}$, $\tilde{s}_1 = [s_1', 0]'$, $\kappa := T_1/(T_1+1)$, and $g \in \mathbb{R}^{p+1}$ is constructed from the incoming observation $(Z_{1,t_{\text{new}}}, Y_{1,t_{\text{new}}})$ as in Proposition 7. Hence, by definition,

$$\begin{aligned} r_1^* =& \dot{\mathbf{Z}}_1^{*\prime}\hat{\mathbf{e}}_1^* = \dot{\mathbf{Z}}_1^{*\prime}\dot{\mathbf{Y}}_1^* - \dot{\mathbf{Z}}_1^{*\prime}\dot{\mathbf{Z}}_1^*\hat{\theta}^* \\ =& (\tilde{s}_1 + \kappa g \dot{y}_{1,t_{\text{new}}}) - (\tilde{S}_1 + \kappa g g')\hat{\theta}^* \\ =& (\tilde{s}_1 + \kappa g \dot{y}_{1,t_{\text{new}}}) - (\tilde{S}_1 + \kappa g g')\tilde{\theta} - (\tilde{S}_1 + \kappa g g')(\hat{\theta}^* - \tilde{\theta}) \\ =& (\tilde{s}_1 - \tilde{S}_1\tilde{\theta}) + \kappa g(\dot{y}_{1,t_{\text{new}}} - g'\tilde{\theta}) - (\tilde{S}_1 + \kappa g g')(\hat{\theta}^* - \tilde{\theta}) \\ =& \tilde{r}_1 + \kappa g \tilde{e}_{1,t_{\text{new}}} - (\tilde{S}_1 + \kappa g g')(\hat{\theta}^* - \tilde{\theta}), \end{aligned} \tag{A.24}$$

where $\tilde{e}_{1,t_{\text{new}}} := \dot{y}_{1,t_{\text{new}}} - g'\tilde{\theta}$. Substituting (A.24) into (A.23) yields a fully expanded exact

update for $\mathbf{M}^*$:

$$\mathbf{M}^* = \widetilde{\mathbf{M}} - \tilde{r}_1\tilde{r}_1' + \left(\tilde{r}_1 + \kappa g\tilde{e}_{1,t_{\text{new}}} - (\tilde{S}_1 + \kappa gg')(\hat{\theta}^* - \tilde{\theta})\right)\left(\tilde{r}_1 + \kappa g\tilde{e}_{1,t_{\text{new}}} - (\tilde{S}_1 + \kappa gg')(\hat{\theta}^* - \tilde{\theta})\right)'.$$